

LYRA: GENERATIVE 3D SCENE RECONSTRUCTION VIA VIDEO DIFFUSION MODEL SELF-DISTILLATION

Anonymous authors
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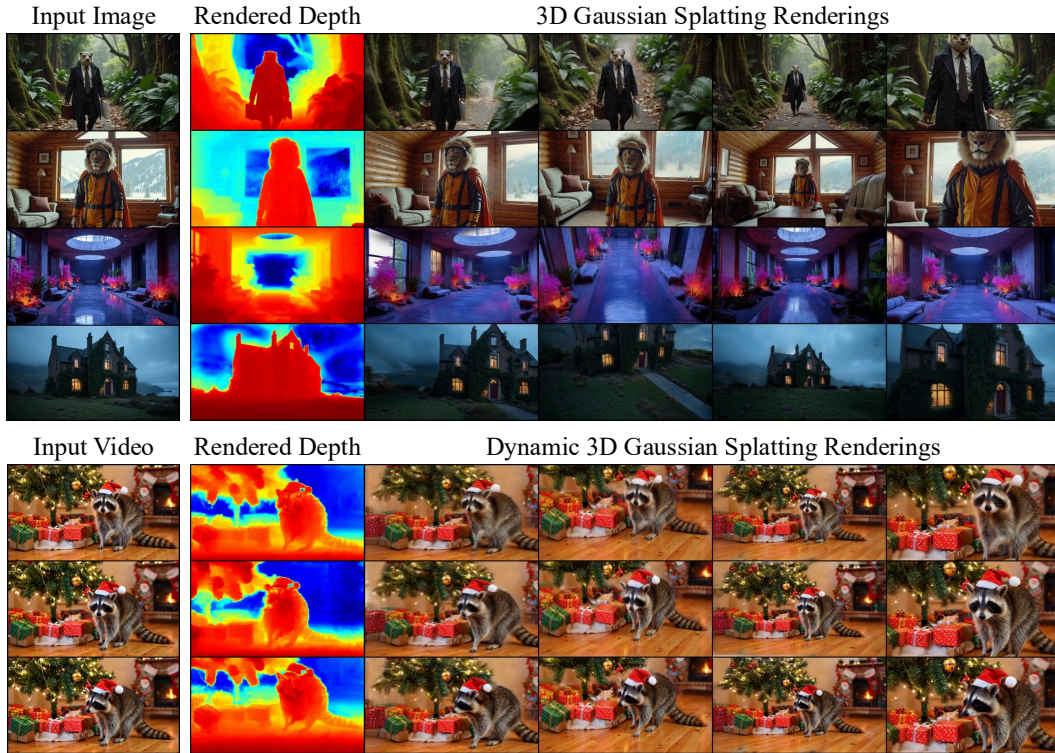


Figure 1: **Feed-Forward 3D and 4D Scene Generation.** From a single image (top), *Lyra* infers a 3D Gaussian Splatting (3DGS) representation in a feed-forward fashion, through self-distilling a video diffusion model without requiring real-world multi-view data. With a video input (bottom), *Lyra* infers a dynamic 3DGS that offers interactive control in both time (rows) and viewpoint (columns).

ABSTRACT

Generating virtual environments is crucial for applications ranging from gaming to physical AI domains such as robotics, autonomous driving, and industrial AI, yet today’s learning-based 3D reconstruction methods depend on captured real-world multi-view data that is rarely available at scale. Video diffusion models offer a tempting alternative because of their remarkable imagination capabilities, but their purely 2D output cannot support simulation where a robot must navigate and interact with an environment. This paper resolves that tension with three contributions. **First**, we introduce a self-distillation framework that distills the implicit 3D knowledge inside a video diffusion model into an explicit 3D Gaussian Splatting (3DGS) representation, removing the need for multi-view training data: concretely, we augment the standard RGB decoder with a 3DGS decoder that is supervised directly by the RGB decoder’s output, so the 3DGS decoder can be trained purely on synthetic data generated by the video diffusion model. **Second**, at inference time this design lets our model synthesize 3D scenes from either a text prompt or a single image and render them in real time, with no per-scene optimization. **Third**, we extend the same framework to dynamic 3D scene generation

from a monocular input video. Across both settings, our experiments show that the framework achieves state-of-the-art performance in static and dynamic 3D scene generation.

1 INTRODUCTION

Creating high-quality 3D environments at scale is a long-standing challenge in computer vision and graphics that underpins applications across film-making, VR/AR, and physical AI with closed-loop simulation. What all of these applications share is a common requirement: explicit 3D representations that support real-time rendering, physical interaction, and consistent multi-view synthesis. This paper targets exactly that requirement, and we organize the introduction around the gap that current methods leave open and the three contributions with which we close it.

Recent advances in neural 3D reconstruction (Mildenhall et al., 2020; Kerbl et al., 2023; Moenne-Loccoz et al., 2024) recover such representations from posed images, but their reliance on accurate camera poses and high-quality images sharply limits scalability. The challenge is even greater for dynamic scenes, which typically require synchronized multi-camera setups (Li et al., 2022). Reconstruction methods are also inherently constrained to the observed content and cannot extrapolate beyond the input views. Feed-forward reconstruction models (Zhang et al., 2024e; Ren et al., 2024c) mitigate some of these limitations, yet they hit a bottleneck of their own: the scarcity of diverse, large-scale 3D training data, which leads to poor out-of-domain generalization. In short, the community lacks a way to obtain explicit, generalizable 3D representations without large multi-view captures.

Video diffusion models (Cosmos, 2025; Wan, 2025) offer a promising escape from this data bottleneck. Trained on massive internet-scale video corpora, they achieve impressive fidelity and generalization across diverse environments: by learning from real-world videos with varied camera trajectories, ranging from handheld motion to cinematic panning and drone footage, they implicitly encode cues about the underlying 3D world without requiring multi-view training data. Unlike reconstruction approaches (Charatan et al., 2024), generative models can moreover hallucinate plausible content beyond what is visible in the input frames. The catch is that video diffusion models generate only 2D frames and lack an explicit 3D representation, which limits their use in simulation and downstream tasks that demand geometric consistency, long-term coherence, and physical interaction. Encouragingly, recent work (Wang et al., 2024e; Ren et al., 2025) shows that video models can be adapted for explicit camera control using relatively small-scale posed datasets. This ability to generate posed image sequences is the key enabler we exploit: it turns a video model into a source that can serve as both input and supervision for a 3D reconstruction model.

Motivated by this insight, we bridge the two paradigms, reconstruction and generation, through what we call *generative 3D scene reconstruction*, and we realize it in *Lyra*, a novel method that generates explicit 3D environments from the latent representations of video diffusion models in a single forward pass. The central idea, and our first contribution, is a self-distillation framework (Fig. 2) in which a 3D Gaussian Splatting (3DGS) decoder operates directly in the latent space of a video diffusion model. Concretely, given a single input image or video, we sample a camera trajectory as conditioning input, denoise the resulting video latent, and decode it along two parallel branches: the first uses the standard RGB decoder to synthesize a video sequence, while the second is our 3DGS decoder, which produces an explicit 3D representation. Together these branches form the self-distillation loop, with the RGB branch (teacher) supervising the 3DGS branch (student). To extend viewpoint coverage, we sample multiple camera trajectories, generate multiple video latents, and train the 3DGS decoder to fuse information across them while mitigating long-term and multi-view inconsistencies.

This framing pays off in three concrete benefits that directly motivate our contributions: (i) it enables generation of large-scale synthetic environments spanning diverse scenarios directly from video diffusion models, removing the need for real-world multi-view captures; (ii) by operating in the latent space of a video model, it processes many views efficiently and avoids the heavy memory overhead of pixel-space feed-forward reconstruction; and (iii) its explicit 3DGS output guarantees geometric consistency, yielding representations that are directly usable in downstream tasks such as physical simulation. As a result, at inference time *Lyra* generates high-quality 3D Gaussian scenes from monocular input that render in real time, *without any additional optimization or post-processing*.

Building on the same design, our second contribution extends the self-distillation framework to dynamic 4D generation from monocular video. Here the video model (teacher) provides space–time

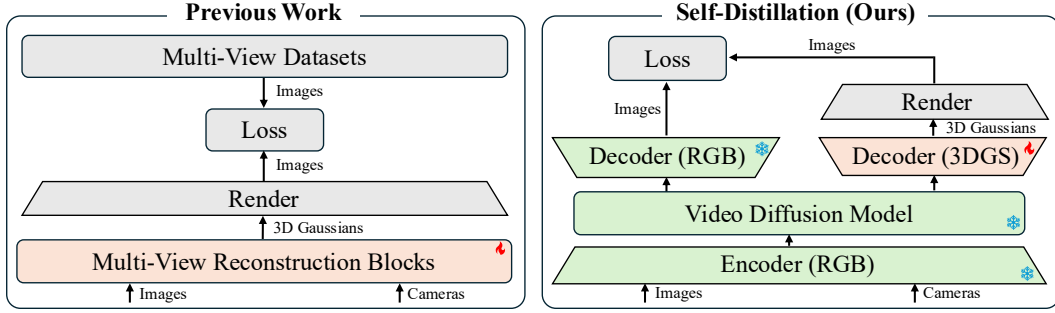


Figure 2: **Self-distillation framework of *Lyra*.** Previous work (left) (Szymanowicz et al., 2025b) trains multi-view reconstruction blocks using real-world datasets with limited diversity (Zhou et al., 2018; Ling et al., 2024b). In contrast, we propose a self-distillation framework (right) for generative 3D scene reconstruction. Precisely, a pre-trained camera-controlled video diffusion model with its RGB decoder output (teacher) supervises the rendering of the 3DGS decoder (student). Using a video model pre-trained on diverse 2D video data allows us to provide diverse multi-view supervision.

supervision, while the student learns to produce time-conditioned 3DGS representations that enable novel-view synthesis of dynamic scenes. Our third contribution is empirical: a single framework that generalizes across diverse scenes and sets the state of the art in both static and dynamic scene generation.

To make the value of the paper easy to track, we summarize our contributions as follows:

- **Self-distillation for 3D reconstruction.** We introduce a self-distillation framework that trains a 3DGS student decoder using a pre-trained camera-controlled video diffusion model as the teacher, eliminating the need for captured multi-view real-world data.
- **Extension to 4D.** We show that the same framework extends with minimal changes to dynamic scenes, supporting 4D reconstruction from a monocular video.
- **State-of-the-art generalization.** We demonstrate that our model generalizes across diverse scenes, achieving state-of-the-art results in single-image 3D scene generation and single-video 4D scene generation.

2 RELATED WORK

Multi-view image generation. Early works on multi-view image generation (Watson et al., 2023; Liu et al., 2023; Shi et al., 2024; Wang & Shi, 2023) mainly focus on object-centric scenes without background. Follow-up works (Sargent et al., 2024; Tang et al., 2023b; Schneider et al., 2025) extend multi-view image generators to the scene scale. CAT3D (Gao et al., 2024b) extends a pre-trained image diffusion model for multi-view image generation from any generated or real image input. In a subsequent stage, the generated multi-view images are reconstructed into an explicit 3D representation, e.g., NeRF (Mildenhall et al., 2020) or 3DGS (Kerbl et al., 2023). Moreover, these methods have been extended to generate dynamic 3D scenes from multiple viewpoints using a space-time diffusion model (Watson et al., 2024; Wu et al., 2025b; Kuang et al., 2024; Wang et al., 2025a). Even though these methods demonstrate impressive results, grounding their generations in an explicit 3D representation requires an expensive per-scene optimization stage that is not amortized across scenes. This is precisely the cost our first contribution removes: instead of creating multi-view images and then reconstructing them through optimization, we *directly* generate a 3D scene from text or a single image in a single forward pass.

Camera-conditioned video generation. There has been significant progress in fine-tuning video diffusion models for 3D camera control. MotionCtrl (Wang et al., 2024e) pioneered camera control by conditioning pre-trained video models with camera poses. Follow-up works (He et al., 2024a; Xu et al., 2024c; Bahmani et al., 2025a;b) represent cameras as Plücker coordinates for pixel-wise conditioning. Another line of work (Hu et al., 2024; Hou et al., 2024; Zhang et al., 2024b) investigates training-free camera control of video diffusion models. Recently, SynCamMaster (Bai et al., 2025b) and ReCamMaster (Bai et al., 2025a) demonstrated impressive camera-controlled video synthesis using large-scale synthetic training data. While these works achieve compelling results, their output is not grounded in 3D. Our contribution is orthogonal and complementary: we tackle feed-forward 3D generation from camera-controlled video models, enabling novel-viewpoint rendering *after* generation. Concretely, we build upon GEN3C (Ren et al., 2025), a recent camera-controlled

video diffusion model, and repurpose it as a teacher within our self-distillation framework for 3D reconstruction.

Feed-forward 3D models. Early work on feed-forward 3D reconstruction mainly focused on object-centric scenes without background. These methods directly predict a NeRF (Hong et al., 2024; Li et al., 2024b) or 3DGS (Tang et al., 2024a; Szymanowicz et al., 2024) from either text or an image. More recent works (Charatan et al., 2024; Zhang et al., 2024e; Szymanowicz et al., 2025a; Ren et al., 2024c) focus on scene-level 3D reconstruction, typically regressing per-pixel 3D Gaussians from one or more input images. However, most scene-level 3D reconstruction methods are limited to training distribution scenes, e.g., RealEstate10K (Zhou et al., 2018), with limited generalizability to generated scenes. Recent methods extend feed-forward 3D reconstruction models to generate scenes. Bolt3D (Szymanowicz et al., 2025b) trains a pointmap (Wang et al., 2024b) autoencoder to generate pointmaps along the multi-view images, used for feed-forward 3D reconstruction. Closely related to our work, Wonderland (Liang et al., 2025a) uses a camera-controlled video model to generate 3D Gaussians with a feed-forward network. Two things set our contributions apart from all of the above. First, we use a camera-controlled video model as a teacher within a self-distillation framework to train our student 3D model *without requiring real-world multi-view data*. Second, we show that this same framework extends with minimal changes to feed-forward 4D scene generation, a task that, to our knowledge, has not been addressed before.

3 SELF-DISTILLATION USING VIDEO DIFFUSION MODELS

This section develops our first contribution: a teacher–student framework that distills the implicit 3D knowledge embedded in video diffusion models into an explicit 3DGS decoder capable of producing high-quality 3D representations. In this framework, the video diffusion model (teacher) generates RGB videos that supervise the 3DGS decoder (student) operating in the same latent space, as illustrated in Fig. 2. We present the framework in the order a reader needs to build it up: we first detail the video diffusion model backbone we rely on (Sec. 3.1) and its role in self-distillation for the 3DGS decoder (Sec. 3.2). We then discuss the design of the 3DGS decoder itself (Sec. 4) and, finally, the minimal changes required to adapt the pipeline to dynamic 3D scenes (Sec. 5), which realizes our second contribution.

3.1 BACKGROUND: CAMERA-CONTROLLED AND 3D-CONSISTENT VIDEO DIFFUSION

Because our 3DGS decoder is supervised solely by the camera-controlled video diffusion model, the quality of our contribution hinges on the teacher’s 3D consistency. For this reason we build on GEN3C (Ren et al., 2025), a recent camera-conditioned video diffusion model, whose key design choices we summarize below to make the rest of the method self-contained.

Spatiotemporal 3D cache. To improve video consistency and camera control precision, GEN3C constructs a spatiotemporal 3D cache $\{\mathbf{P}^{t,v}\}$ from input image(s) or videos, where each $\mathbf{P}^{t,v}$ is a colored point cloud obtained by unprojecting the depth estimation (Wang et al., 2024a) of an RGB image captured from camera viewpoint v at time t . The cache is organized as an $L \times V$ array, where L denotes the number of frames (temporal length) and V denotes the number of camera views.

Rendering and structured guidance. To leverage this cache, GEN3C renders each point cloud from arbitrary given camera poses, producing RGB images $\mathbf{I}^{t,v}$ and disocclusion masks $\mathbf{M}^{t,v}$ via $(\mathbf{I}^{t,v}, \mathbf{M}^{t,v}) = \mathcal{R}(\mathbf{P}^{t,v}, \mathbf{C}^t)$, where $\mathcal{R}$ denotes the rendering function that projects the 3D point cloud $\mathbf{P}^{t,v}$ onto the 2D camera plane according to the camera pose $\mathbf{C}^t$ at time t . Given a sequence of camera poses $\mathbf{C} = (\mathbf{C}^1, \dots, \mathbf{C}^L)$, rendering all cache elements produces V videos of length L , which we stack into image sequences $\mathbf{I}^v \in \mathbb{R}^{L \times 3 \times H \times W}$ and mask sequences $\mathbf{M}^v \in \mathbb{R}^{L \times 1 \times H \times W}$ for each view v . The disocclusion masks flag the regions the video diffusion model must fill in, so that these renderings together act as structured visual guidance for the subsequent video generation.

Video variational autoencoder (VAE). Diffusion-based video generation models commonly employ a video variational autoencoder (VAE) (Kingma & Welling, 2013; Rombach et al., 2022) to compress videos into a lower-dimensional latent space for efficient training and inference. This latent space is not merely an implementation detail for us; it is the space in which our 3DGS decoder operates, and thus central to the efficiency benefit claimed in the introduction. Given an RGB video $\mathbf{I} \in \mathbb{R}^{L \times 3 \times H \times W}$, a pre-trained VAE encoder $\mathcal{E}$ encodes the video into a latent space, i.e., $\mathbf{z} = \mathcal{E}(\mathbf{I}) \in \mathbb{R}^{L' \times C \times h \times w}$, in which both training and inference of the diffusion model are performed. The final

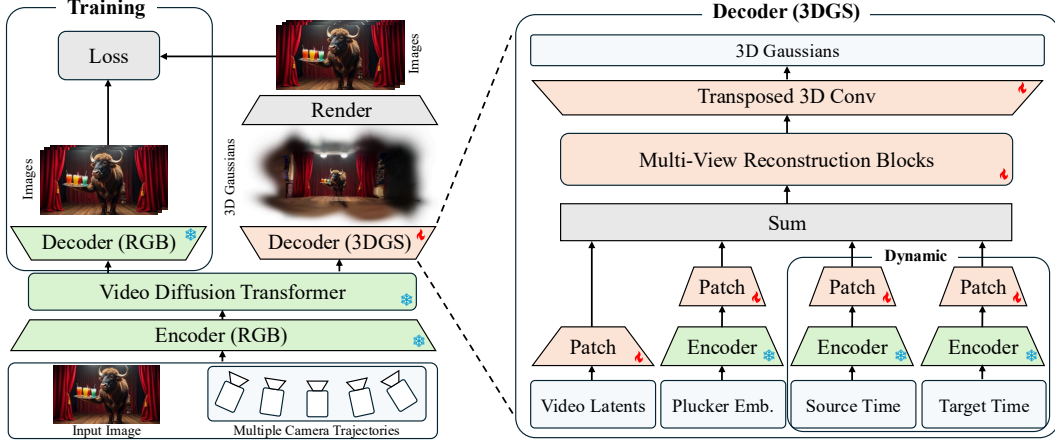


Figure 4: **3D Generative Reconstruction Framework.** Our pipeline builds upon a camera-controlled video diffusion model (Ren et al., 2025) pre-trained on large scale data. We train a 3D Gaussian Splatting (3DGS) decoder by aligning the 2D image renderings of generated 3DGS scenes with the RGB-decoded generations of the pre-trained video model. We only train the 3DGS decoder while freezing the pre-trained autoencoder and diffusion model. At inference time we directly use the 3DGS decoder, without requiring the RGB decoder anymore. Time conditioning within the 3DGS decoder allows us to easily extend our approach from static to dynamic 3D scene generation.

video $\hat{\mathbf{I}} = \mathcal{D}_{rgb}(\mathbf{z})$ is recovered with a pre-trained VAE decoder $\mathcal{D}_{rgb}$. Throughout, we adopt the pre-trained GEN3C (Ren et al., 2025; Cosmos, 2025) model: the latent channel dimension is $C = 16$, the temporal dimension is $L' = (L - 1)/\tau + 1$, and the spatial dimensions are $h = H/\sigma$ and $w = W/\sigma$, with temporal compression factor $\tau = 8$ and spatial compression factor $\sigma = 8$.

3.2 SELF-DISTILLATION

We now describe the core of our first contribution: training the 3DGS decoder $\mathcal{D}_s$ as a student under a teacher–student paradigm in which a camera-controlled video diffusion model $\mathcal{V}$ serves as the teacher. To obtain diverse supervision without any real multi-view capture, we curate a large-scale set of diverse text prompts with large language models (OpenAI, 2025; Bai et al., 2023), generate images I with an image diffusion model (Black Forest Labs, 2024), and expand each into multi-view sequences using GEN3C (Ren et al., 2025).

Teacher–student setup. Given an input image I and a sampled camera trajectory $\{\mathbf{C}^t\}_{t=1}^L$, the video diffusion model $\mathcal{V}$ generates a denoised video latent $\mathbf{z} = \mathcal{V}(I, \{\mathbf{C}^t\}_{t=1}^L)$. The latent $\mathbf{z}$ is decoded along two branches: the pre-trained RGB decoder $\mathcal{D}_{rgb}$ produces video frames $\mathbf{I}_{\mathcal{D}_{rgb}} = \mathcal{D}_{rgb}(\mathbf{z})$, while the 3DGS decoder $\mathcal{D}_s$ outputs explicit 3D Gaussians $\mathbf{G}$. Rendered views from 3DGS (*student*) are defined as $\mathbf{I}_{\mathcal{D}_s} = \text{Render}(\mathbf{G}, \{\mathbf{C}^t\}_{t=1}^L)$ and are supervised to match RGB frames $\mathbf{I}_{\mathcal{D}_{rgb}}$ (*teacher*), forming the self-distillation loop.

Multi-trajectory supervision. To enlarge viewpoint coverage for the input image, we sample $V = 6$ camera trajectories per input image, as shown in Fig. 3. For each trajectory v , we construct a spatiotemporal cache $\{\mathbf{P}^{t,v}\}_{t=1}^L$ along with $L = 121$ camera poses $\{\mathbf{C}^{t,v}\}_{t=1}^L$. Passing these caches through the video diffusion model $\mathcal{V}$ yields latents $\mathbf{z}^v$, which the teacher decodes into RGB frames $\mathbf{I}_{\mathcal{D}_{rgb}}^v = \mathcal{D}_{rgb}(\mathbf{z}^v)$. The 3DGS decoder $\mathcal{D}_s$ then learns to fuse these multiple $\mathbf{z}^v$ into coherent Gaussians $\mathbf{G}$ while filling disoccluded regions. This design delivers the three benefits promised in the introduction: the decoder is trained *entirely* from synthetic supervision provided by $\mathcal{V}$ (no real multi-view data), operating in latent space enables *efficient* aggregation of multiple trajectories, and the explicit Gaussian output $\mathbf{G}$ enforces *geometric consistency* that supports downstream simulation and real-time rendering.

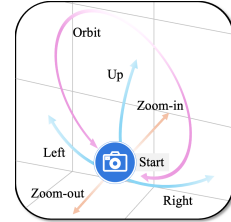


Figure 3: Sampled six camera trajectories to maximize view coverage.

4 FEED-FORWARD RECONSTRUCTION FROM MULTI-VIEW VIDEO LATENTS

Having established the self-distillation loop, we now turn to the decoder that makes it scalable, which is the technical crux of our efficiency benefit. Our feed-forward 3DGS decoder $\mathcal{D}_s$ transforms the synthesized multi-view latents generated by our video model $\mathcal{V}$ into an explicit 3D representation that can be rendered from arbitrary viewpoints. We adopt 3D Gaussian Splatting (3DGS) (Kerbl et al., 2023) for its explicit representation and fast rendering speed, and we detail the decoder in Fig. 4 (right).

4.1 3DGS DECODER

Scaling Latent-Based 3D Reconstruction. The design choice that unlocks our scale is to reconstruct in latent space rather than pixel space. Previous feed-forward reconstruction frameworks generate 3D Gaussians for each pixel and are therefore limited by the number and resolution of input views they can handle: GS-LRM (Zhang et al., 2024e) operates on 2–4 images at 512×512 resolution, and AnySplat (Jiang et al., 2025a) is trained with 24 images at 448×448 resolution. By contrast, our video model synthesizes $V \times L = 6 \times 121 = 726$ input views at 704×1280 resolution—far beyond the capacity of these methods and well past the GPU memory limit of a pixel-space decoder. The bottleneck is the attention mechanism over visual tokens, whose memory and compute grow with the number of pixels. Our remedy is to avoid scaling in pixel space altogether and instead operate directly in the compressed video latent space produced by our camera-controlled video diffusion model, denoting the multi-view video latents as $\mathbf{Z} \in \mathbb{R}^{V \times L' \times C \times h \times w}$.

Architecture. The 3DGS decoder $\mathcal{D}_s$ maps its inputs into the hidden dimension of the main reconstruction blocks and outputs per-pixel 3D Gaussian features $\mathbf{G} \in \mathbb{R}^{V \times L \times H \times W \times 14}$. The inputs are the multi-view video latents $\mathbf{Z}$ and encoded Plücker embeddings $\mathbf{E}$, both with latent dimensions $\mathbf{Z}, \mathbf{E} \in \mathbb{R}^{V \times L' \times C \times h \times w}$. A deliberately lightweight front end keeps the decoder efficient: each input component is patchified to the hidden dimension of the reconstruction blocks, and we require *no* additional visual encoder—only a spatial 2×2 patchification layer (Dosovitskiy et al., 2021) to turn the video latents into flattened tokens. The sum of both inputs is then processed through the reconstruction blocks. We follow the design introduced in Long-LRM (Ziwen et al., 2024): one block consists of one Transformer (Vaswani et al., 2017) layer followed by seven Mamba-2 (Dao & Gu, 2024) layers, and we repeat the block twice to obtain 16 layers with 512 hidden dimensions. Finally, a transposed 3D convolution maps the hidden representation to 14 Gaussian channels: 3D position (x, y, z) , scale (s_x, s_y, s_z) , rotation quaternion (q_w, q_x, q_y, q_z) , opacity α , and RGB (r, g, b) . Formally, the static decoder is $\mathbf{G} = \mathcal{D}_s(\mathbf{Z}, \mathbf{E})$.

Plücker embeddings. To inject camera information without a bespoke encoder, we again reuse existing components. Raw Plücker embeddings are computed from camera poses $\{\mathbf{C}^{t,v}\}$ and intrinsics as $\mathbf{E}^{\text{raw}} \in \mathbb{R}^{V \times L \times 6 \times H \times W}$, and we encode them with the RGB encoder $\mathcal{E}$ of the pre-trained VAE. Concretely, we separately encode the 3-dimensional ray directions and the 3-dimensional cross product of ray directions and origins using $\mathcal{E}$, then concatenate the latents along the channel dimension to obtain $\mathbf{E}^{\text{enc}} \in \mathbb{R}^{V \times L' \times 2C \times h \times w}$. A lightweight MLP maps $\mathbf{E}^{\text{enc}}$ to $\mathbf{E} \in \mathbb{R}^{V \times L' \times C \times h \times w}$, halving the channel dimension to match the video latent $\mathbf{Z}$.

4.2 LOSS FUNCTION

Since the teacher supplies only image-space supervision, our loss design is what turns that signal into consistent 3D geometry; each term below is later validated in our ablation study.

Image-based supervision. The primary signal is an image-based reconstruction loss, split into a Mean Squared Error (MSE) loss $\mathcal{L}_{mse}$ and an LPIPS loss $\mathcal{L}_{lips}$ (Zhang et al., 2018) that uses VGG (Simonyan & Zisserman, 2014) as a feature extractor.

Depth supervision. We found that a decoder trained with RGB loss alone often produces flattened geometry. To fix this, we additionally supervise the rendered depth maps using the consistent video depth estimated by the off-the-shelf system ViPE (Huang et al., 2025), together with the scale-invariant depth loss $\mathcal{L}_{depth}$ from Long-LRM (Ziwen et al., 2024).

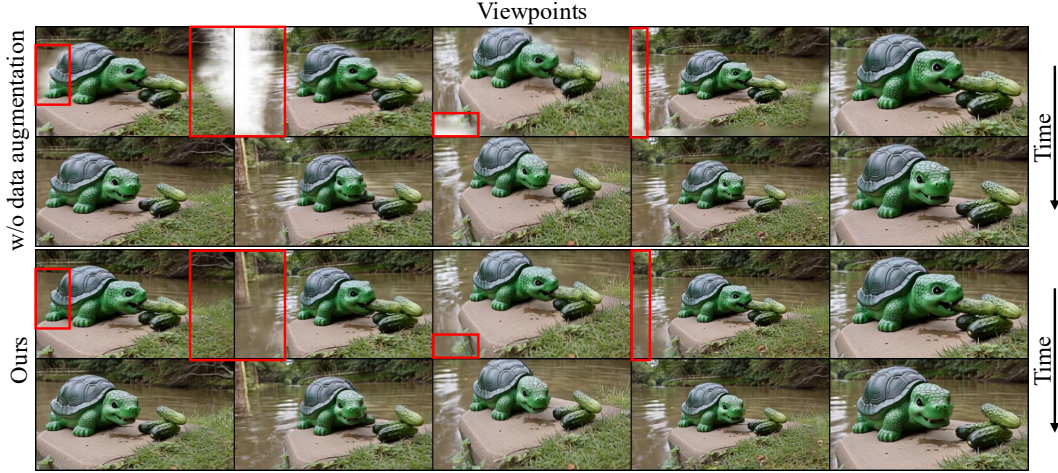


Figure 5: **Dynamic data augmentation.** When naively supervising the time-aware 3DGS decoder, we observe artifacts in the generated 3D Gaussians. Specifically, early timesteps from extreme poses exhibit low opacity in regions not covered by the supervision signal. To address this, we augment the supervision data with a motion-reversed video, ensuring that each timestep is observed from the full spatial coverage, thereby preventing low opacity artifacts in the early timesteps.

Opacity-based pruning. Following Long-LRM (Ziwen et al., 2024), we apply an L1 regularization on the opacity $\mathcal{L}_{opacity}$ and remove the Gaussians with the lowest 80% opacity, keeping the representation compact.

Total loss. Our total loss is computed as the weighted sum of all losses with weight factors λ_i

$$\mathcal{L} = \lambda_{mse} \mathcal{L}_{mse} + \lambda_{lpips} \mathcal{L}_{lpips} + \lambda_{depth} \mathcal{L}_{depth} + \lambda_{opacity} \mathcal{L}_{opacity} \quad (1)$$

where we set $\lambda_{mse} = 1.0$, $\lambda_{lpips} = 0.5$, $\lambda_{depth} = 0.05$, and $\lambda_{opacity} = 0.1$.

5 EXTENSION TO DYNAMIC 3D SCENES

We now present our second contribution: extending the framework to dynamic 3D scenes. A key selling point is that this requires only minimal changes to the static pipeline, which we outline below.

Self-distillation for dynamic scenes. Our dynamic 3D setup deliberately mirrors the static counterpart, so that the same self-distillation contribution carries over. The only change to the input is that, instead of a single image, the video model takes a single video of length L with corresponding camera poses and generates multi-view video latents capturing the same underlying motion. For video inputs, we follow a protocol similar to the static case: sampling diverse text prompts with large language models, then generating videos using video diffusion models (Cosmos, 2025; Wan, 2025). The generated videos are then annotated with camera poses and depth maps using ViPE (Huang et al., 2025).

Dynamic 3DGS decoder architecture. Consistent with our claim of minimal changes, the dynamic decoder reuses the static architecture almost verbatim. We follow the bullet-time design of Liang et al. (2025b) and let our dynamic 3DGS decoder $\mathcal{D}_s$ generate time-dependent 3D Gaussians for all video frames. Specifically, $\mathcal{D}_s$ closely follows the architecture of $\mathcal{D}_s$, except that we augment its input with the encoded source and target time embeddings $\mathbf{T}^{src}, \mathbf{T}^{tgt} \in \mathbb{R}^{V \times L' \times C \times h \times w}$. The unencoded (raw) source times are assigned to each input frame, forming $\mathbf{T}^{src, raw} \in \mathbb{R}^{V \times L \times 1 \times H \times W}$, and the target time is $\mathbf{T}^{tgt, raw} \in \mathbb{R}^{V \times 1 \times 1 \times H \times W}$. Both are spatially repeated and augmented with a 2-dimensional sinusoidal embedding to expand the channel dimension to 3, then encoded with the RGB encoder $\mathcal{E}$ to latent dimensions $\mathbf{T}^{src}, \mathbf{T}^{tgt} \in \mathbb{R}^{V \times L' \times C \times h \times w}$. The target time is repeated along the latent temporal dimension to match L' . During training, a target time step is randomly sampled, and the corresponding 3D Gaussians are decoded and supervised across all trajectories generated with dynamic data augmentation. We fine-tune $\mathcal{D}_d$ from a pre-trained $\mathcal{D}_s$ with the patchification layers of $\mathbf{T}^{src}$ and $\mathbf{T}^{tgt}$ initialized to zeros. $\mathbf{T}^{src}$ and $\mathbf{T}^{tgt}$ are directly added to the sum of $\mathbf{Z}$ and $\mathbf{E}$. Formally, the dynamic decoder is defined as $\mathbf{G} = \mathcal{D}_d(\mathbf{Z}, \mathbf{E}, \mathbf{T}^{src}, \mathbf{T}^{tgt})$.

Table 1: **State-of-the-art comparisons.** Evidence for our state-of-the-art contribution: on single-image-to-3D generation across RealEstate10K, DL3DV, and Tanks-and-Temples, our method (bottom row) leads every prior work on every metric.

Method	RealEstate10K			DL3DV			Tanks-and-Temples		
	PSNR $\uparrow$	SSIM $\uparrow$	LPIPS $\downarrow$	PSNR $\uparrow$	SSIM $\uparrow$	LPIPS $\downarrow$	PSNR $\uparrow$	SSIM $\uparrow$	LPIPS $\downarrow$
ZeroNVS	13.01	0.378	0.448	13.35	0.339	0.465	12.94	0.325	0.470
ViewCrafter	16.84	0.514	0.341	15.53	0.525	0.352	14.93	0.483	0.384
Wonderland	17.15	0.550	0.292	16.64	0.574	0.325	15.90	0.510	0.344
Bolt3D	21.54	0.747	0.234	-	-	-	-	-	-
Ours	21.79	0.752	0.219	20.09	0.583	0.313	19.24	0.570	0.336

Dynamic data augmentation. Extending self-distillation to time introduces one failure mode that we must address for the second contribution to work in practice. In static scenes, where time is not a factor, frames from different timesteps can all supervise the 3DGS renderings. In dynamic scenes, however, the 3DGS changes over time, so only frames from the corresponding timestamp are valid supervision. This restriction invites a trivial solution in which the 3DGS at a given timestep simply ignores information from frames at other timesteps (see Fig. 5).

Our fix, and the one component unique to the dynamic setting, is a dynamic data augmentation strategy that balances supervision across timesteps. Early frames are naturally associated with viewpoints close to the input image, whereas later frames correspond to more distant viewpoints. To counter this imbalance, we augment the training data with paired supervision views for each timestep — one near and one far. Concretely, we reverse the input video in frame order and feed it into the video model $\mathcal{V}$, which produces six additional multi-view sequences. After reversing these sequences back to the original motion order, we obtain six trajectories that progress inward (from far viewpoints toward the input). Combined with the original six outward trajectories, this yields 12 supervision views per timestep. Importantly, this augmentation is applied only during training: the reversed trajectories act purely as extra supervision signals to prevent collapse from pruning artifacts and are not required at inference time.

6 EXPERIMENTS

6.1 EXPERIMENTAL SETUP

Datasets. Consistent with our first contribution, we use *no* existing multi-view datasets for training; instead, we construct our own, which we call the *Lyra* dataset, relying solely on our video model to supervise the 3DGS decoder. The 3D reconstruction setup uses 59,031 images and the 4D setup 7,378 videos, all derived from diverse text prompts spanning indoor and outdoor environments, humans, animals, and both realistic and imaginative content. Synthesizing six camera trajectories for each image (3D) or video (4D) yields 354,186 videos for 3D and 44,268 videos for 4D — a scale that would be impractical to capture in the real world, underscoring the data-free nature of our approach.

Baselines. To substantiate our third contribution, we compare against the state-of-the-art approaches for generative single-image-to-3D reconstruction: ZeroNVS (Sargent et al., 2024), ViewCrafter (Yu et al., 2024b), Wonderland (Liang et al., 2025a), and Bolt3D (Szymanowicz et al., 2025b). Because *no source code is available* to run these methods on our out-of-distribution evaluation set, we rely primarily on the quantitative comparisons reported in their papers.

Evaluation. Following previous works, we evaluate single-image-to-3D on RealEstate10K (Zhou et al., 2018), DL3DV (Ling et al., 2024b), and Tanks and Temples (Knapitsch et al., 2017), adopting the evaluation protocol of Wonderland (Liang et al., 2025a) and Bolt3D (Szymanowicz et al., 2025b) and reporting the standard reconstruction metrics PSNR, SSIM, and LPIPS.

6.2 MAIN RESULTS

Quantitative results. Table 1 reports single-image-to-3D results and directly supports our state-of-the-art claim: *Lyra* outperforms all prior works on every benchmark across every metric. A notable consequence for the field is that, because our reconstructions inherit the quality of the teacher, the



Figure 6: **Image-to-3DGS Generation.** We visualize five views from generated 3DGS scenes.



Figure 7: **Ablations.** We visualize ablation results by rendering the same extreme novel viewpoint after image-to-3DGS generation; depth visualizations and ablations are provided in Appendix C.4.

clearest path to further gains is simply a stronger video generative model rather than a redesigned reconstructor. Additional quantitative comparisons on the *Lyra* dataset are provided in Appendix C.

Qualitative results. Figures 1 and 6 show novel-view renderings from our generated 3DGS scenes; video results will be released upon acceptance. These results illustrate the same contribution qualitatively: our method generates high-quality novel-view content for both 3D and 4D scenes in unseen regions while staying consistent with the input image or video. Further qualitative comparisons on the *Lyra* dataset appear in Appendix C.

6.3 ABLATIONS

The ablations in Tab. 2 verify that each design choice behind our contributions actually earns its place, evaluated on out-of-distribution diverse prompts by comparing 3DGS renderings against the videos generated by our camera-controlled video diffusion model. We walk through the components below, grouped by data, loss, and architecture.

Real data only. Replacing self-distillation with training on only real multi-view datasets, i.e., RealEstate10K (Zhou et al., 2018) and DL3DV (Ling et al., 2024b), sharply degrades quality. Although these datasets are standard in prior work, they generalize poorly to out-of-distribution scenes, which is exactly the weakness our self-distillation contribution is designed to overcome.

Self-distillation + real data. Perhaps surprisingly, jointly training with self-distillation *and* real data does not improve over self-distillation alone. This is a strong endorsement of our first contribution: the synthetic supervision is already diverse and consistent enough to learn a reconstruction model on its own, and adding real captures buys nothing.

No depth loss. The depth loss prevents flat geometry and even slightly improves image-based metrics, validating this term of our loss design. We visualize depth maps in Appendix C.4.

Table 2: **Ablation study on the *Lyra* dataset.** Each row removes one design choice behind our contributions; the full model (top) is best or tied-best throughout.

Method	PSNR $\uparrow$	SSIM $\uparrow$	LPIPS $\downarrow$
Ours	24.77	0.837	0.224
Data			
real data only	19.08	0.659	0.413
self-distill. + real data	24.74	0.823	0.236
Loss			
w/o depth loss	24.31	0.811	0.247
w/o opacity pruning	24.55	0.820	0.237
9 w/o LPIPS loss	23.74	0.766	0.370
Architecture			
w/o multi-view fusion	17.73	0.632	0.446
w/o Mamba-2	24.58	0.818	0.241

No opacity pruning. Keeping all Gaussians instead of pruning them by opacity is both heavier and slightly worse: learning to prune most Gaussians makes the representation more compact and marginally improves visual quality. Concretely, rendering at $H = 704$, $W = 1280$ takes 18 ms with pruning versus 30 ms without, a $1.67\times$ speedup.

No LPIPS loss. The LPIPS loss improves robustness to input inconsistencies and better preserves high-frequency details, again confirming a component of our loss design.

No multi-view fusion. A naive alternative would generate 3D Gaussians for each camera trajectory independently and then merge them into one point cloud. We instead learn multi-view fusion across trajectories within our reconstruction blocks, where each token attends to the others, and observe a large advantage — direct evidence for the fusion design at the heart of our decoder.

No Mamba-2. Replacing our joint Transformer/Mamba-2 blocks with Transformer-only blocks slightly lowers quality while being far slower: one forward pass with $V = 6$, $L = 121$, $H = 704$, $W = 1280$ takes 3213 ms with joint blocks versus 20922 ms with Transformer-only blocks, a $6.5\times$ speedup.

No latent-based 3DGS. Finally, this ablation isolates the design choice that enables our scale. Pixel-space methods such as BTimer (Liang et al., 2025b) are restricted to 12 input frames, whereas we handle up to 726; consequently, running the 3DGS decoder in pixel space instead of latent space runs out of memory outright.

7 CONCLUSION

We presented *Lyra*, a 3D and 4D generation framework that requires only a single image or video as input, and we delivered on three contributions. **First**, instead of collecting multi-view datasets, we introduced camera-controlled video diffusion models as teachers within a self-distillation framework for a student 3DGS decoder that operates in the video model’s latent space and directly reconstructs 3D Gaussians without any post-processing or optimization. **Second**, we showed that this same framework extends with minimal changes to dynamic scenes, establishing the feasibility of 4D generation from monocular video. **Third**, we demonstrated that this design generalizes across diverse scenes and attains state-of-the-art results in both static and dynamic settings. A natural limitation follows from the first contribution: the scale and consistency of our generated scenes are bounded by the capacity of the camera-controlled video diffusion model, so improving that teacher will directly enable larger-scale, more consistent scene synthesis. Two further directions are promising: adapting auto-regressive techniques (Chen et al., 2024a) into our framework for large-scale generation, and modeling motion and tracking information within the reconstruction network, as in concurrent work (Lin et al., 2025), to improve visual motion quality.

8 ETHICS STATEMENT

Like all generative AI technologies, our method carries risks of misuse, including producing misleading or synthetic 3D/4D content. While such risks exist, the goal of this research is to contribute to simulation, robotics, and embodied AI by providing scalable, controllable tools for data generation. In this spirit, we expect the broader impact of our contributions to be realized through the following capabilities and insights.

- **The ability to enable realistic simulation for embodied agents:** We provide controllable and physically consistent camera motion within generated 3D and 4D environments to support training and evaluating agents that must perceive, navigate, and interact with complex scenes.
- **Improved scalability for data engines:** Our method removes the need for multi-view capture and expensive per-scene optimization, and helps produce high-fidelity, customizable scenes at scale, which is particularly suitable for research in robotics, reinforcement learning, and closed-loop simulation.

- **Better technical understanding:** As an academic study, this work contributes a foundational case study of how camera control interacts with spatiotemporal generative models, which will benefit research in computer vision, graphics, and embodied AI.

While we caution against deceptive or wrongful use of this technology—where misleading and/or unverifiable media could be created with realistic generative outputs—we believe that the adoption of safeguards (e.g., provenance tracking, dataset documentation, and good evaluation practices) will be useful in ensuring that generative models advance science and society in positive ways.

9 REPRODUCIBILITY STATEMENT

To make all three contributions fully reproducible and to accelerate future research, we will release our training and inference code, model weights, and data for both 3D and 4D generation upon acceptance.

REFERENCES

- Sherwin Bahmani, Jeong Joon Park, Despoina Paschalidou, Hao Tang, Gordon Wetzstein, Leonidas Guibas, Luc Van Gool, and Radu Timofte. 3d-aware video generation. In *TMLR*, 2023a.
- Sherwin Bahmani, Jeong Joon Park, Despoina Paschalidou, Xingguang Yan, Gordon Wetzstein, Leonidas Guibas, and Andrea Tagliasacchi. CC3D: Layout-conditioned generation of compositional 3D scenes. In *Proc. ICCV*, 2023b.
- Sherwin Bahmani, Xian Liu, Wang Yifan, Ivan Skorokhodov, Victor Rong, Ziwei Liu, Xihui Liu, Jeong Joon Park, Sergey Tulyakov, Gordon Wetzstein, Andrea Tagliasacchi, and David B. Lindell. Tc4d: Trajectory-conditioned text-to-4d generation. In *Proc. ECCV*, 2024a.
- Sherwin Bahmani, Ivan Skorokhodov, Victor Rong, Gordon Wetzstein, Leonidas Guibas, Peter Wonka, Sergey Tulyakov, Jeong Joon Park, Andrea Tagliasacchi, and David B. Lindell. 4d-fy: Text-to-4d generation using hybrid score distillation sampling. In *Proc. CVPR*, 2024b.
- Sherwin Bahmani, Ivan Skorokhodov, Guocheng Qian, Aliaksandr Siarohin, Willi Menapace, Andrea Tagliasacchi, David B Lindell, and Sergey Tulyakov. Ac3d: Analyzing and improving 3d camera control in video diffusion transformers. *Proc. CVPR*, 2025a.
- Sherwin Bahmani, Ivan Skorokhodov, Aliaksandr Siarohin, Willi Menapace, Guocheng Qian, Michael Vasilkovsky, Hsin-Ying Lee, Chaoyang Wang, Jiaxu Zou, Andrea Tagliasacchi, et al. Vd3d: Taming large video diffusion transformers for 3d camera control. *Proc. ICLR*, 2025b.
- Jianhong Bai, Menghan Xia, Xiao Fu, Xintao Wang, Lianrui Mu, Jinwen Cao, Zuozhu Liu, Haoji Hu, Xiang Bai, Pengfei Wan, et al. Recammaster: Camera-controlled generative rendering from a single video. *arXiv preprint arXiv:2503.11647*, 2025a.
- Jianhong Bai, Menghan Xia, Xintao Wang, Ziyang Yuan, Xiao Fu, Zuozhu Liu, Haoji Hu, Pengfei Wan, and Di Zhang. Syncammaster: Synchronizing multi-camera video generation from diverse viewpoints. *Proc. ICLR*, 2025b.
- Jinze Bai, Shuai Bai, Shusheng Yang, Shijie Wang, Sinan Tan, Peng Wang, Junyang Lin, Chang Zhou, and Jingren Zhou. Qwen-vl: A versatile vision-language model for understanding, localization, text reading, and beyond. *arXiv preprint arXiv:2308.12966*, 2023.
- Black Forest Labs. Flux. <https://github.com/black-forest-labs/flux>, 2024.
- Andreas Blattmann, Tim Dockhorn, Sumith Kulal, Daniel Mendelevitch, Maciej Kilian, Dominik Lorenz, Yam Levi, Zion English, Vikram Voleti, Adam Letts, et al. Stable video diffusion: Scaling latent video diffusion models to large datasets. *arXiv preprint arXiv:2311.15127*, 2023.
- Chenjie Cao, Jingkai Zhou, Shikai Li, Jingyun Liang, Chaohui Yu, Fan Wang, Xiangyang Xue, and Yanwei Fu. Uni3c: Unifying precisely 3d-enhanced camera and human motion controls for video generation. *arXiv preprint arXiv:2504.14899*, 2025.

594 Yukang Cao, Liang Pan, Kai Han, Kwan-Yee K Wong, and Ziwei Liu. Avatargo: Zero-shot 4d
595 human-object interaction generation and animation. *arXiv preprint arXiv:2410.07164*, 2024.

596

597 Zenghao Chai, Chen Tang, Yongkang Wong, and Mohan Kankanhalli. Star: Skeleton-aware text-
598 based 4d avatar generation with in-network motion retargeting. *arXiv preprint arXiv:2406.04629*,
599 2024.

600

601 Eric R Chan, Connor Z Lin, Matthew A Chan, Koki Nagano, Boxiao Pan, Shalini De Mello, Orazio
602 Gallo, Leonidas J Guibas, Jonathan Tremblay, Sameh Khamis, et al. Efficient geometry-aware 3D
603 generative adversarial networks. In *Proc. CVPR*, 2022.

604

605 Eric R Chan, Koki Nagano, Matthew A Chan, Alexander W Bergman, Jeong Joon Park, Axel Levy,
606 Miika Aittala, Shalini De Mello, Tero Karras, and Gordon Wetzstein. Generative novel view
synthesis with 3d-aware diffusion models. In *Proc. ICCV*, 2023.

607

608 David Charatan, Sizhe Lester Li, Andrea Tagliasacchi, and Vincent Sitzmann. pixelsplat: 3d gaussian
609 splats from image pairs for scalable generalizable 3d reconstruction. In *Proc. CVPR*, 2024.

610

611 Boyuan Chen, Diego Martí Monsó, Yilun Du, Max Simchowitz, Russ Tedrake, and Vincent Sitzmann.
Diffusion forcing: Next-token prediction meets full-sequence diffusion. *Proc. NeurIPS*, 2024a.

612

613 Ce Chen, Shaoli Huang, Xuelin Chen, Guangyi Chen, Xiaoguang Han, Kun Zhang, and Ming-
614 ming Gong. Ct4d: Consistent text-to-4d generation with animatable meshes. *arXiv preprint*
arXiv:2408.08342, 2024b.

615

616 Haoxin Chen, Menghan Xia, Yingqing He, Yong Zhang, Xiaodong Cun, Shaoshu Yang, Jinbo Xing,
617 Yaofang Liu, Qifeng Chen, Xintao Wang, Chao Weng, and Ying Shan. Videocrafter1: Open
618 diffusion models for high-quality video generation. *arXiv preprint arXiv:2310.19512*, 2023a.

619

620 Kevin Chen, Christopher B Choy, Manolis Savva, Angel X Chang, Thomas Funkhouser, and Silvio
621 Savarese. Text2Shape: Generating shapes from natural language by learning joint embeddings. In
Proc. ACCV, 2018.

622

623 Rui Chen, Yongwei Chen, Ningxin Jiao, and Kui Jia. Fantasia3D: Disentangling geometry and
624 appearance for high-quality text-to-3D content creation. *arXiv preprint arXiv:2303.13873*, 2023b.

625

626 Soon Yau Cheong, Duygu Ceylan, Armin Mustafa, Andrew Gilbert, and Chun-Hao Paul Huang.
627 Boosting camera motion control for video diffusion transformers. *arXiv preprint arXiv:2410.10802*,
628 2024.

629

630 Wen-Hsuan Chu, Lei Ke, and Katerina Fragkiadaki. Dreamscene4d: Dynamic multi-object scene
generation from monocular videos. *arXiv preprint arXiv:2405.02280*, 2024.

631

632 Team Cosmos. Cosmos world foundation model platform for physical ai. *arXiv preprint*
arXiv:2501.03575, 2025.

633

634 Tri Dao and Albert Gu. Transformers are ssms: Generalized models and efficient algorithms through
635 structured state space duality. *arXiv preprint arXiv:2405.21060*, 2024.

636

637 Terrance DeVries, Miguel Angel Bautista, Nitish Srivastava, Graham W Taylor, and Joshua M
638 Susskind. Unconstrained scene generation with locally conditioned radiance fields. In *Proc. ICCV*,
639 2021.

640

641 Alexey Dosovitskiy, Lucas Beyer, Alexander Kolesnikov, Dirk Weissenborn, Xiaohua Zhai, Thomas
642 Unterthiner, Mostafa Dehghani, Matthias Minderer, Georg Heigold, Sylvain Gelly, et al. An image
is worth 16x16 words: Transformers for image recognition at scale. In *Proc. ICLR*, 2021.

643

644 Qijun Feng, Zhen Xing, Zuxuan Wu, and Yu-Gang Jiang. FDGaussian: Fast Gaussian splatting from
645 single image via geometric-aware diffusion model. *arXiv preprint arXiv:2403.10242*, 2024a.

646

647 Yutao Feng, Yintong Shang, Xiang Feng, Lei Lan, Shandian Zhe, Tianjia Shao, Hongzhi Wu, Kun
Zhou, Hao Su, Chenfanfu Jiang, et al. Elastogen: 4d generative elastodynamics. *arXiv preprint*
arXiv:2405.15056, 2024b.

648 Quankai Gao, Qiangeng Xu, Zhe Cao, Ben Mildenhall, Wenchao Ma, Le Chen, Danhang Tang, and
649 Ulrich Neumann. Gaussianflow: Splatting Gaussian dynamics for 4D content creation. *arXiv*
650 *preprint arXiv:2403.12365*, 2024a.

651

652 Ruiqi Gao, Aleksander Holynski, Philipp Henzler, Arthur Brussee, Ricardo Martin-Brualla, Pratul
653 Srinivasan, Jonathan T Barron, and Ben Poole. Cat3d: Create anything in 3d with multi-view
654 diffusion models. In *Proc. NeurIPS*, 2024b.

655

656 William Gao, Noam Aigerman, Thibault Groueix, Vova Kim, and Rana Hanocka. TextDeformer:
657 Geometry manipulation using text guidance. In *SIGGRAPH*, 2023.

658

659 Jiatao Gu, Alex Trevithick, Kai-En Lin, Joshua M Susskind, Christian Theobalt, Lingjie Liu, and
660 Ravi Ramamoorthi. NerfDiff: Single-image view synthesis with NeRF-guided distillation from
661 3D-aware diffusion. In *Proc. ICML*, 2023.

662

663 Zekai Gu, Rui Yan, Jiahao Lu, Peng Li, Zhiyang Dou, Chenyang Si, Zhen Dong, Qifeng Liu, Cheng
664 Lin, Ziwei Liu, et al. Diffusion as shader: 3d-aware video diffusion for versatile video generation
665 control. In *SIGGRAPH*, 2025.

666

667 Junlin Han, Filippos Kokkinos, and Philip Torr. VFusion3D: Learning scalable 3D generative models
668 from video diffusion models. *arXiv preprint arXiv:2403.12034*, 2024.

669

670 Hao He, Yinghao Xu, Yuwei Guo, Gordon Wetzstein, Bo Dai, Hongsheng Li, and Ceyuan Yang. Cam-
671 eractrl: Enabling camera control for text-to-video generation. *arXiv preprint arXiv:2404.02101*,
672 2024a.

673

674 Hao He, Ceyuan Yang, Shanchuan Lin, Yinghao Xu, Meng Wei, Liangke Gui, Qi Zhao, Gordon
675 Wetzstein, Lu Jiang, and Hongsheng Li. Cameractrl ii: Dynamic scene exploration via camera-
676 controlled video diffusion models. *arXiv preprint arXiv:2503.10592*, 2025.

677

678 Xianglong He, Junyi Chen, Sida Peng, Di Huang, Yangguang Li, Xiaoshui Huang, Chun Yuan, Wanli
679 Ouyang, and Tong He. GVGEn: Text-to-3D generation with volumetric representation. *arXiv*
680 *preprint arXiv:2403.12957*, 2024b.

681

682 Lukas Höllein, Ang Cao, Andrew Owens, Justin Johnson, and Matthias Nießner. Text2room:
683 Extracting textured 3d meshes from 2d text-to-image models. In *Proc. ICCV*, 2023.

684

685 Lukas Höllein, Aljaž Božič, Norman Müller, David Novotny, Hung-Yu Tseng, Christian Richardt,
686 Michael Zollhöfer, and Matthias Nießner. ViewDiff: 3D-consistent image generation with text-to-
687 image models. In *Proc. CVPR*, 2024.

688

689 Yicong Hong, Kai Zhang, Jiuxiang Gu, Sai Bi, Yang Zhou, Difan Liu, Feng Liu, Kalyan Sunkavalli,
690 Trung Bui, and Hao Tan. LRM: Large reconstruction model for single image to 3D. In *Proc. ICLR*,
691 2024.

692

693 Chen Hou, Guoqiang Wei, Yan Zeng, and Zhibo Chen. Training-free camera control for video
694 generation. *arXiv preprint arXiv:2406.10126*, 2024.

695

696 Ronghang Hu, Nikhila Ravi, Alexander C Berg, and Deepak Pathak. Worldsheet: Wrapping the
697 world in a 3d sheet for view synthesis from a single image. In *Proc. ICCV*, 2021.

698

699 Tao Hu, Haoyang Peng, Xiao Liu, and Yuewen Ma. Ex-4d: Extreme viewpoint 4d video synthesis
700 via depth watertight mesh. *arXiv preprint arXiv:2506.05554*, 2025.

701

Teng Hu, Jiangning Zhang, Ran Yi, Yating Wang, Hongrui Huang, Jieyu Weng, Yabiao Wang, and
Lizhuang Ma. Motionmaster: Training-free camera motion transfer for video generation. *arXiv*
preprint arXiv:2404.15789, 2024.

Chun-Hao Paul Huang, Jae Shin Yoon, Hyeonho Jeong, Niloy Mitra, and Duygu Ceylan. Camera
pose estimation emerging in video diffusion transformer, 2024a. URL <https://openreview.net/forum?id=lgf2LW7fOJ>.

-
- Jiahui Huang, Qunjie Zhou, Hesam Rabeti, Aleksandr Korovko, Huan Ling, Xuanchi Ren, Tianchang Shen, Jun Gao, Dmitry Slepichev, Chen-Hsuan Lin, Jiawei Ren, Kevin Xie, Joydeep Biswas, Laura Leal-Taixe, and Sanja Fidler. Vipe: Video pose engine for 3d geometric perception. In *NVIDIA Research Whitepapers*, 2025.
- Tianyu Huang, Yihan Zeng, Hui Li, Wangmeng Zuo, and Rynson WH Lau. Dreamphysics: Learning physical properties of dynamic 3d gaussians with video diffusion priors. *arXiv preprint arXiv:2406.01476*, 2024b.
- Ajay Jain, Ben Mildenhall, Jonathan T Barron, Pieter Abbeel, and Ben Poole. Zero-shot text-guided object generation with dream fields. In *Proc. CVPR*, 2022.
- Nikolay Jetchev. ClipMatrix: Text-controlled creation of 3D textured meshes. *arXiv preprint arXiv:2109.12922*, 2021.
- Lihan Jiang, Yucheng Mao, Linning Xu, Tao Lu, Kerui Ren, Yichen Jin, Xudong Xu, Mulin Yu, Jiangmiao Pang, Feng Zhao, et al. Anysplat: Feed-forward 3d gaussian splatting from unconstrained views. *arXiv preprint arXiv:2505.23716*, 2025a.
- Lutao Jiang and Lin Wang. Brightdreamer: Generic 3D Gaussian generative framework for fast text-to-3D synthesis. *arXiv preprint arXiv:2403.11273*, 2024.
- Yanqin Jiang, Chaohui Yu, Chenjie Cao, Fan Wang, Weiming Hu, and Jin Gao. Animate3d: Animating any 3d model with multi-view video diffusion. *arXiv preprint arXiv:2407.11398*, 2024a.
- Yanqin Jiang, Li Zhang, Jin Gao, Weimin Hu, and Yao Yao. Consistent4d: Consistent 360 $\{\deg\}$ dynamic object generation from monocular video. In *Proc. ICLR*, 2024b.
- Zeren Jiang, Chuanxia Zheng, Iro Laina, Diane Larlus, and Andrea Vedaldi. Geo4d: Leveraging video generators for geometric 4d scene reconstruction. *arXiv preprint arXiv:2504.07961*, 2025b.
- Wonjoon Jin, Qi Dai, Chong Luo, Seung-Hwan Baek, and Sunghyun Cho. Flovd: Optical flow meets video diffusion model for enhanced camera-controlled video synthesis. In *Proc. CVPR*, 2025.
- Xuan Ju, Weicai Ye, Quande Liu, Qiulin Wang, Xintao Wang, Pengfei Wan, Di Zhang, Kun Gai, and Qiang Xu. Fulldit: Multi-task video generative foundation model with full attention. *arXiv preprint arXiv:2503.19907*, 2025.
- Yash Kant, Aliaksandr Siarohin, Ziyi Wu, Michael Vasilkovsky, Guocheng Qian, Jian Ren, Riza Alp Guler, Bernard Ghanem, Sergey Tulyakov, and Igor Gilitschenski. Spad: Spatially aware multi-view diffusers. In *Proc. CVPR*, 2024.
- Yash Kant, Ethan Weber, Jin Kyu Kim, Rawal Khrodar, Su Zhaoen, Julieta Martinez, Igor Gilitschenski, Shunsuke Saito, and Timur Bagautdinov. Pippo: High-resolution multi-view humans from a single image. In *Proc. CVPR*, 2025.
- Oren Katzir, Or Patashnik, Daniel Cohen-Or, and Dani Lischinski. Noise-free score distillation. In *Proc. ICLR*, 2024.
- Bernhard Kerbl, Georgios Kopanas, Thomas Leimkühler, and George Drettakis. 3d gaussian splatting for real-time radiance field rendering. In *ACM TOG*, 2023.
- Seung Wook Kim, Bradley Brown, Kangxue Yin, Karsten Kreis, Katja Schwarz, Daiqing Li, Robin Rombach, Antonio Torralba, and Sanja Fidler. NeuralField-LDM: Scene generation with hierarchical latent diffusion models. In *Proc. CVPR*, 2023.
- Diederik P Kingma and Max Welling. Auto-encoding variational bayes. *arXiv preprint arXiv:1312.6114*, 2013.
- Mert Kiray, Paul Uhlenbruck, Nassir Navab, and Benjamin Busam. Promptvfx: Text-driven fields for open-world 3d gaussian animation. *arXiv preprint arXiv:2506.01091*, 2025.
- Tobias Kirschstein, Javier Romero, Artem Sevastopolsky, Matthias Nießner, and Shunsuke Saito. Avat3r: Large animatable gaussian reconstruction model for high-fidelity 3d head avatars. *arXiv preprint arXiv:2502.20220*, 2025.

-
- Arno Knapitsch, Jaesik Park, Qian-Yi Zhou, and Vladlen Koltun. Tanks and temples: Benchmarking large-scale scene reconstruction. *ACM TOG*, 2017.
- Zhengfei Kuang, Shengqu Cai, Hao He, Yinghao Xu, Hongsheng Li, Leonidas Guibas, and Gordon Wetzstein. Collaborative video diffusion: Consistent multi-video generation with camera control. In *Proc. NeurIPS*, 2024.
- Kyungmin Lee, Kihyuk Sohn, and Jinwoo Shin. DreamFlow: High-quality text-to-3D generation by approximating probability flow. In *Proc. ICLR*, 2024a.
- Yao-Chih Lee, Yi-Ting Chen, Andrew Wang, Ting-Hsuan Liao, Brandon Y Feng, and Jia-Bin Huang. Vividdream: Generating 3d scene with ambient dynamics. *arXiv preprint arXiv:2405.20334*, 2024b.
- Guojun Lei, Chi Wang, Hong Li, Rong Zhang, Yikai Wang, and Weiwei Xu. Animateanything: Consistent and controllable animation for video generation. *arXiv preprint arXiv:2411.10836*, 2024.
- Guojun Lei, Chi Wang, Rong Zhang, Yikai Wang, Hong Li, and Weiwei Xu. Animateanything: Consistent and controllable animation for video generation. In *Proc. CVPR*, 2025.
- Bing Li, Cheng Zheng, Wenxuan Zhu, Jinjie Mai, Biao Zhang, Peter Wonka, and Bernard Ghanem. Vivid-zoo: Multi-view video generation with diffusion model. *arXiv preprint arXiv:2406.08659*, 2024a.
- Jiahao Li, Hao Tan, Kai Zhang, Zexiang Xu, Fujun Luan, Yinghao Xu, Yicong Hong, Kalyan Sunkavalli, Greg Shakhnarovich, and Sai Bi. Instant3D: Fast text-to-3D with sparse-view generation and large reconstruction model. In *Proc. ICLR*, 2024b.
- Jinwei Li, Huan-ang Gao, Wenyi Li, Haohan Chi, Chenyu Liu, Chenxi Du, Yiqian Liu, Mingju Gao, Guiyu Zhang, Zongzheng Zhang, et al. Fb-4d: Spatial-temporal coherent dynamic 3d content generation with feature banks. *arXiv preprint arXiv:2503.20784*, 2025a.
- Longfei Li, Zhiwen Fan, Wenyan Cong, Xinhang Liu, Yuyang Yin, Matt Foutter, Panwang Pan, Chenyu You, Yue Wang, Zhangyang Wang, et al. Martian world models: Controllable video synthesis with physically accurate 3d reconstructions. *arXiv preprint arXiv:2507.07978*, 2025b.
- Renjie Li, Panwang Pan, Bangbang Yang, Dejia Xu, Shijie Zhou, Xuanyang Zhang, Zeming Li, Achuta Kadambi, Zhangyang Wang, and Zhiwen Fan. 4k4dgen: Panoramic 4d generation at 4k resolution. *arXiv preprint arXiv:2406.13527*, 2024c.
- Teng Li, Guangcong Zheng, Rui Jiang, Shuigen Zhan, Tao Wu, Yehao Lu, Yining Lin, Chuanyun Deng, Yapan Xiong, Min Chen, et al. Realcam-i2v: Real-world image-to-video generation with interactive complex camera control. *arXiv preprint arXiv:2502.10059*, 2025c.
- Tianye Li, Mira Slavcheva, Michael Zollhoefer, Simon Green, Christoph Lassner, Changil Kim, Tanner Schmidt, Steven Lovegrove, Michael Goesele, Richard Newcombe, et al. Neural 3d video synthesis from multi-view video. In *Proc. CVPR*, 2022.
- Zhiqi Li, Yiming Chen, and Peidong Liu. Dreammesh4d: Video-to-4d generation with sparse-controlled gaussian-mesh hybrid representation. *arXiv preprint arXiv:2410.06756*, 2024d.
- Zhiqi Li, Yiming Chen, Lingzhe Zhao, and Peidong Liu. Controllable text-to-3D generation via surface-aligned Gaussian splatting. *arXiv preprint arXiv:2403.09981*, 2024e.
- Hanwen Liang, Yuyang Yin, Dejia Xu, Hanxue Liang, Zhangyang Wang, Konstantinos N Platanotis, Yao Zhao, and Yunchao Wei. Diffusion4d: Fast spatial-temporal consistent 4d generation via video diffusion models. *arXiv preprint arXiv:2405.16645*, 2024.
- Hanwen Liang, Junli Cao, Vidit Goel, Guocheng Qian, Sergei Korolev, Demetri Terzopoulos, Konstantinos N Platanotis, Sergey Tulyakov, and Jian Ren. Wonderland: Navigating 3d scenes from a single image. *Proc. CVPR*, 2025a.

-
- Hanxue Liang, Jiawei Ren, Ashkan Mirzaei, Antonio Torralba, Ziwei Liu, Igor Gilitschenski, Sanja Fidler, Cengiz Oztireli, Huan Ling, Zan Gojcic, and Jiahui Huang. Feed-forward bullet-time reconstruction of dynamic scenes from monocular videos. *Proc. NeurIPS*, 2025b.
- Yixun Liang, Xin Yang, Jiantao Lin, Haodong Li, Xiaogang Xu, and Yingcong Chen. Lucid-dreamer: Towards high-fidelity text-to-3D generation via interval score matching. *arXiv preprint arXiv:2311.11284*, 2023.
- Chen-Hsuan Lin, Jun Gao, Luming Tang, Towaki Takikawa, Xiaohui Zeng, Xun Huang, Karsten Kreis, Sanja Fidler, Ming-Yu Liu, and Tsung-Yi Lin. Magic3D: High-resolution text-to-3D content creation. In *Proc. CVPR*, 2023a.
- Chenguo Lin, Yuchen Lin, Panwang Pan, Yifan Yu, Honglei Yan, Katerina Fragkiadaki, and Yadong Mu. Movies: Motion-aware 4d dynamic view synthesis in one second. *arXiv preprint arXiv:2507.10065*, 2025.
- Jiajing Lin, Zhenzhong Wang, Yongjie Hou, Yuzhou Tang, and Min Jiang. Phy124: Fast physics-driven 4d content generation from a single image. *arXiv preprint arXiv:2409.07179*, 2024.
- Yukang Lin, Haonan Han, Chaoqun Gong, Zunnan Xu, Yachao Zhang, and Xiu Li. Consistent123: One image to highly consistent 3D asset using case-aware diffusion priors. In *arXiv preprint arXiv:2309.17261*, 2023b.
- Huan Ling, Seung Wook Kim, Antonio Torralba, Sanja Fidler, and Karsten Kreis. Align your gaussians: Text-to-4d with dynamic 3d gaussians and composed diffusion models. In *Proc. CVPR*, 2024a.
- Lu Ling, Yichen Sheng, Zhi Tu, Wentian Zhao, Cheng Xin, Kun Wan, Lantao Yu, Qianyu Guo, Zixun Yu, Yawen Lu, et al. D13dv-10k: A large-scale scene dataset for deep learning-based 3d vision. In *Proc. CVPR*, 2024b.
- Pengyang Ling, Jiazi Bu, Pan Zhang, Xiaoyi Dong, Yuhang Zang, Tong Wu, Huaian Chen, Jiaqi Wang, and Yi Jin. Motionclone: Training-free motion cloning for controllable video generation. *arXiv preprint arXiv:2406.05338*, 2024c.
- Lijuan Liu, Wenfa Li, Dongbo Zhang, Shuo Wang, and Shaohui Jiao. Idcnet: Guided video diffusion for metric-consistent rgb-d scene generation with precise camera control. *arXiv preprint arXiv:2508.04147*, 2025.
- Pengkun Liu, Yikai Wang, Fuchun Sun, Jiafang Li, Hang Xiao, Hongxiang Xue, and Xinzhou Wang. Isotropic3D: Image-to-3D generation based on a single clip embedding. *arXiv preprint arXiv:2403.10395*, 2024a.
- Ruoshi Liu, Rundi Wu, Basile Van Hoorick, Pavel Tokmakov, Sergey Zakharov, and Carl Vondrick. Zero-1-to-3: Zero-shot one image to 3D object. In *Proc. ICCV*, 2023.
- Xian Liu, Xiaohang Zhan, Jiayang Tang, Ying Shan, Gang Zeng, Dahua Lin, Xihui Liu, and Ziwei Liu. HumanGaussian: Text-driven 3D human generation with Gaussian splatting. In *Proc. CVPR*, 2024b.
- Yuan Liu, Cheng Lin, Zijiao Zeng, Xiaoxiao Long, Lingjie Liu, Taku Komura, and Wenping Wang. SyncDreamer: Generating multiview-consistent images from a single-view image. In *Proc. ICLR*, 2024c.
- Xiaoxiao Long, Yuan-Chen Guo, Cheng Lin, Yuan Liu, Zhiyang Dou, Lingjie Liu, Yuexin Ma, Song-Hai Zhang, Marc Habermann, Christian Theobalt, et al. Wonder3D: Single image to 3D using cross-domain diffusion. In *Proc. CVPR*, 2024.
- Yifan Lu, Xuanchi Ren, Jiawei Yang, Tianchang Shen, Zhangjie Wu, Jun Gao, Yue Wang, Siheng Chen, Mike Chen, Sanja Fidler, et al. Infinicube: Unbounded and controllable dynamic 3d driving scene generation with world-guided video models. *arXiv preprint arXiv:2412.03934*, 2024.

864 Yue Ma, Kunyu Feng, Xinhua Zhang, Hongyu Liu, David Junhao Zhang, Jinbo Xing, Yinhan Zhang,
865 Ayden Yang, Zeyu Wang, and Qifeng Chen. Follow-your-creation: Empowering 4d creation
866 through video inpainting. *arXiv preprint arXiv:2506.04590*, 2025.

867 Jinjie Mai, Wenxuan Zhu, Haozhe Liu, Bing Li, Cheng Zheng, Jürgen Schmidhuber, and Bernard
868 Ghanem. Can video diffusion model reconstruct 4d geometry? *arXiv preprint arXiv:2503.21082*,
869 2025.

870 Qiaowei Miao, Yawei Luo, and Yi Yang. Pla4d: Pixel-level alignments for text-to-4d gaussian
871 splatting. *arXiv preprint arXiv:2405.19957*, 2024.

872 Ben Mildenhall, Pratul P Srinivasan, Matthew Tancik, Jonathan T Barron, Ravi Ramamoorthi, and
873 Ren Ng. Nerf: Representing scenes as neural radiance fields for view synthesis. In *Proc. ECCV*,
874 2020.

875 Nicolas Moenne-Loccoz, Ashkan Mirzaei, Or Perel, Riccardo de Lutio, Janick Martinez Esturo,
876 Gavriel State, Sanja Fidler, Nicholas Sharp, and Zan Gojcic. 3d gaussian ray tracing: Fast tracing
877 of particle scenes. *ACM Transactions on Graphics and SIGGRAPH Asia*, 2024.

878 OpenAI. Chatgpt (gpt-5), 2025. Large language model. Available at <https://chat.openai.com/>.

879 Roy Or-El, Xuan Luo, Mengyi Shan, Eli Shechtman, Jeong Joon Park, and Ira Kemelmacher-
880 Shlizerman. StyleSDF: High-resolution 3D-consistent image and geometry generation. In *Proc.*
881 *CVPR*, 2022.

882 Zijie Pan, Zeyu Yang, Xiatian Zhu, and Li Zhang. Fast dynamic 3d object generation from a
883 single-view video. *arXiv preprint arXiv:2401.08742*, 2024.

884 Jangho Park, Taesung Kwon, and Jong Chul Ye. Zero4d: Training-free 4d video generation from
885 single video using off-the-shelf video diffusion. *arXiv preprint arXiv:2503.22622*, 2025.

886 Ben Poole, Ajay Jain, Jonathan T. Barron, and Ben Mildenhall. DreamFusion: Text-to-3D using 2D
887 diffusion. In *Proc. ICLR*, 2023.

888 Guocheng Qian, Junli Cao, Aliaksandr Siarohin, Yash Kant, Chaoyang Wang, Michael Vasilkovsky,
889 Hsin-Ying Lee, Yuwei Fang, Ivan Skorokhodov, Peiye Zhuang, et al. Atom: Amortized text-to-
890 mesh using 2d diffusion. *arXiv preprint arXiv:2402.00867*, 2024a.

891 Guocheng Qian, Jinjie Mai, Abdullah Hamdi, Jian Ren, Aliaksandr Siarohin, Bing Li, Hsin-Ying Lee,
892 Ivan Skorokhodov, Peter Wonka, Sergey Tulyakov, et al. Magic123: One image to high-quality 3D
893 object generation using both 2D and 3D diffusion priors. In *Proc. ICLR*, 2024b.

894 Alec Radford, Jong Wook Kim, Chris Hallacy, Aditya Ramesh, Gabriel Goh, Sandhini Agarwal,
895 Girish Sastry, Amanda Askell, Pamela Mishkin, Jack Clark, et al. Learning transferable visual
896 models from natural language supervision. In *Proc. ICML*, 2021.

897 Jiawei Ren, Liang Pan, Jiaxiang Tang, Chi Zhang, Ang Cao, Gang Zeng, and Ziwei Liu. DreamGaus-
898 sian4D: Generative 4D Gaussian splatting. *arXiv preprint arXiv:2312.17142*, 2023.

899 Jiawei Ren, Kevin Xie, Ashkan Mirzaei, Hanxue Liang, Xiaohui Zeng, Karsten Kreis, Ziwei Liu,
900 Antonio Torralba, Sanja Fidler, Seung Wook Kim, et al. L4gm: Large 4d gaussian reconstruction
901 model. In *Proc. NeurIPS*, 2024a.

902 Xuanchi Ren, Jiahui Huang, Xiaohui Zeng, Ken Museth, Sanja Fidler, and Francis Williams. Xcube:
903 Large-scale 3d generative modeling using sparse voxel hierarchies. In *Proc. CVPR*, 2024b.

904 Xuanchi Ren, Yifan Lu, Hanxue Liang, Zhangjie Wu, Huan Ling, Mike Chen, Sanja Fidler, Francis
905 Williams, and Jiahui Huang. Scube: Instant large-scale scene reconstruction using voxplats. *Proc.*
906 *NeurIPS*, 2024c.

907 Xuanchi Ren, Tianchang Shen, Jiahui Huang, Huan Ling, Yifan Lu, Merlin Nimier-David, Thomas
908 Müller, Alexander Keller, Sanja Fidler, and Jun Gao. Gen3c: 3d-informed world-consistent video
909 generation with precise camera control. *Proc. CVPR*, 2025.

-
- Robin Rombach, Andreas Blattmann, Dominik Lorenz, Patrick Esser, and Björn Ommer. High-resolution image synthesis with latent diffusion models. In *Proc. CVPR*, 2022.
- Aditya Sanghi, Hang Chu, Joseph G Lambourne, Ye Wang, Chin-Yi Cheng, Marco Fumero, and Kamal Rahimi Malekshan. CLIP-Forge: Towards zero-shot text-to-shape generation. In *Proc. CVPR*, 2022.
- Kyle Sargent, Zizhang Li, Tanmay Shah, Charles Herrmann, Hong-Xing Yu, Yunzhi Zhang, Eric Ryan Chan, Dmitry Lagun, Li Fei-Fei, Deqing Sun, et al. Zeronvs: Zero-shot 360-degree view synthesis from a single image. In *Proc. CVPR*, 2024.
- Manuel-Andreas Schneider, Lukas Höllein, and Matthias Nießner. Worldexplorer: Towards generating fully navigable 3d scenes. *arXiv preprint arXiv:2506.01799*, 2025.
- Katja Schwarz, Axel Sauer, Michael Niemeyer, Yiyi Liao, and Andreas Geiger. VoxGRAF: Fast 3D-aware image synthesis with sparse voxel grids. In *Proc. NeurIPS*, 2022.
- Katja Schwarz, Norman Mueller, and Peter Kotschieder. Generative gaussian splatting: Generating 3d scenes with video diffusion priors. *arXiv preprint arXiv:2503.13272*, 2025.
- Junyoung Seo, Jisang Han, Jaewoo Jung, Siyoon Jin, Joungbin Lee, Takuya Narihira, Kazumi Fukuda, Takashi Shibuya, Donghoon Ahn, Shoukang Hu, et al. Vid-camedit: Video camera trajectory editing with generative rendering from estimated geometry. *arXiv preprint arXiv:2506.13697*, 2025.
- Yichun Shi, Peng Wang, Jianglong Ye, Long Mai, Kejie Li, and Xiao Yang. MVDream: Multi-view diffusion for 3D generation. In *Proc. ICLR*, 2024.
- Jaidev Shriram, Alex Trevithick, Lingjie Liu, and Ravi Ramamoorthi. Realmdreamer: Text-driven 3d scene generation with inpainting and depth diffusion. *arXiv preprint arXiv:2404.07199*, 2024.
- Karen Simonyan and Andrew Zisserman. Very deep convolutional networks for large-scale image recognition. *arXiv preprint arXiv:1409.1556*, 2014.
- Uriel Singer, Shelly Sheynin, Adam Polyak, Oron Ashual, Iurii Makarov, Filippos Kokkinos, Naman Goyal, Andrea Vedaldi, Devi Parikh, Justin Johnson, et al. Text-to-4d dynamic scene generation. In *Proc. ICML*, 2023.
- Jingxiang Sun, Bo Zhang, Ruizhi Shao, Lizhen Wang, Wen Liu, Zhenda Xie, and Yebin Liu. DreamCraft3D: Hierarchical 3D generation with bootstrapped diffusion prior. In *Proc. ICLR*, 2024a.
- Keqiang Sun, Dor Litvak, Yunzhi Zhang, Hongsheng Li, Jiajun Wu, and Shangzhe Wu. Ponymation: Learning articulated 3d animal motions from unlabeled online videos. In *Proc. ECCV*, 2024b.
- Qi Sun, Zhiyang Guo, Ziyu Wan, Jing Nathan Yan, Shengming Yin, Wengang Zhou, Jing Liao, and Houqiang Li. Eg4d: Explicit generation of 4d object without score distillation. *arXiv preprint arXiv:2405.18132*, 2024c.
- Wenqiang Sun, Shuo Chen, Fangfu Liu, Zilong Chen, Yueqi Duan, Jun Zhang, and Yikai Wang. Dimensionx: Create any 3d and 4d scenes from a single image with controllable video diffusion. *arXiv preprint arXiv:2411.04928*, 2024d.
- Stanislaw Szymanowicz, Christian Rupprecht, and Andrea Vedaldi. Viewset diffusion:(0-) image-conditioned 3d generative models from 2d data. In *Proc. ICCV*, 2023.
- Stanislaw Szymanowicz, Christian Rupprecht, and Andrea Vedaldi. Splatler image: Ultra-fast single-view 3D reconstruction. In *Proc. CVPR*, 2024.
- Stanislaw Szymanowicz, Eldar Insafutdinov, Chuanxia Zheng, Dylan Campbell, João F Henriques, Christian Rupprecht, and Andrea Vedaldi. Flash3d: Feed-forward generalisable 3d scene reconstruction from a single image. *Proc. 3DV*, 2025a.

Stanislaw Szymanowicz, Jason Y Zhang, Pratul Srinivasan, Ruiqi Gao, Arthur Brussee, Aleksander Holynski, Ricardo Martin-Brualla, Jonathan T Barron, and Philipp Henzler. Bolt3d: Generating 3d scenes in seconds. *arXiv preprint arXiv:2503.14445*, 2025b.

Jiaxiang Tang, Zhaoxi Chen, Xiaokang Chen, Tengfei Wang, Gang Zeng, and Ziwei Liu. LGM: Large multi-view gaussian model for high-resolution 3d content creation. In *Proc. ECCV*, 2024a.

Jiaxiang Tang, Zhaoxi Chen, Xiaokang Chen, Tengfei Wang, Gang Zeng, and Ziwei Liu. LGM: Large multi-view gaussian model for high-resolution 3d content creation. *Proc. ECCV*, 2024b.

Junshu Tang, Tengfei Wang, Bo Zhang, Ting Zhang, Ran Yi, Lizhuang Ma, and Dong Chen. Make-it-3D: High-fidelity 3D creation from a single image with diffusion prior. *arXiv preprint arXiv:2303.14184*, 2023a.

Shitao Tang, Fuyang Zhang, Jiacheng Chen, Peng Wang, and Yasutaka Furukawa. Mvdifffusion: Enabling holistic multi-view image generation with correspondence-aware diffusion. *Proc. NeurIPS*, 2023b.

Zhenggang Tang, Peiye Zhuang, Chaoyang Wang, Aliaksandr Siarohin, Yash Kant, Alexander Schwing, Sergey Tulyakov, and Hsin-Ying Lee. Pixel-aligned multi-view generation with depth guided decoder. *arXiv preprint arXiv:2408.14016*, 2024c.

Ayush Tewari, Tianwei Yin, George Cazenavette, Semon Rezhchikov, Josh Tenenbaum, Frédo Durand, Bill Freeman, and Vincent Sitzmann. Diffusion with forward models: Solving stochastic inverse problems without direct supervision. In *Proc. NeurIPS*, 2023.

Dmitry Tochilkin, David Pankratz, Zexiang Liu, Zixuan Huang, Adam Letts, Yangguang Li, Ding Liang, Christian Laforte, Varun Jampani, and Yan-Pei Cao. Triposr: Fast 3D object reconstruction from a single image. *arXiv preprint arXiv:2403.02151*, 2024.

Lukas Uzolas, Elmar Eisemann, and Petr Kellnhofer. Motiondreamer: Zero-shot 3d mesh animation from video diffusion models. *arXiv preprint arXiv:2405.20155*, 2024.

Basile Van Hoorick, Rundi Wu, Ege Ozguroglu, Kyle Sargent, Ruoshi Liu, Pavel Tokmakov, Achal Dave, Changxi Zheng, and Carl Vondrick. Generative camera dolly: Extreme monocular dynamic novel view synthesis. In *Proc. ECCV*, 2024.

Ashish Vaswani, Noam Shazeer, Niki Parmar, Jakob Uszkoreit, Llion Jones, Aidan N Gomez, Łukasz Kaiser, and Illia Polosukhin. Attention is all you need. In *Proc. NeurIPS*, 2017.

Vikram Voleti, Chun-Han Yao, Mark Boss, Adam Letts, David Pankratz, Dmitry Tochilkin, Christian Laforte, Robin Rombach, and Varun Jampani. SV3D: Novel multi-view synthesis and 3D generation from a single image using latent video diffusion. *arXiv preprint arXiv:2403.12008*, 2024.

Team Wan. Wan: Open and advanced large-scale video generative models. *arXiv preprint arXiv:2503.20314*, 2025.

Ziyu Wan, Despoina Paschalidou, Ian Huang, Hongyu Liu, Bokui Shen, Xiaoyu Xiang, Jing Liao, and Leonidas Guibas. CAD: Photorealistic 3D generation via adversarial distillation. In *Proc. CVPR*, 2024.

Can Wang, Menglei Chai, Mingming He, Dongdong Chen, and Jing Liao. Clip-NeRF: Text-and-image driven manipulation of neural radiance fields. In *Proc. CVPR*, 2022.

Chaoyang Wang, Peiye Zhuang, Tuan Duc Ngo, Willi Menapace, Aliaksandr Siarohin, Michael Vasilkovsky, Ivan Skorokhodov, Sergey Tulyakov, Peter Wonka, and Hsin-Ying Lee. 4real-video: Learning generalizable photo-realistic 4d video diffusion. *Proc. CVPR*, 2025a.

Haochen Wang, Xiaodan Du, Jiahao Li, Raymond A Yeh, and Greg Shakhnarovich. Score Jacobian chaining: Lifting pretrained 2d diffusion models for 3D generation. In *Proc. CVPR*, 2023a.

Peng Wang and Yichun Shi. Imagedream: Image-prompt multi-view diffusion for 3d generation. *arXiv preprint arXiv:2312.02201*, 2023.

1026 Qinghe Wang, Yawen Luo, Xiaoyu Shi, Xu Jia, Huchuan Lu, Tianfan Xue, Xintao Wang, Pengfei
1027 Wan, Di Zhang, and Kun Gai. Cinemaster: A 3d-aware and controllable framework for cinematic
1028 text-to-video generation. In *SIGGRAPH*, 2025b.

1029 Ruicheng Wang, Sicheng Xu, Cassie Dai, Jianfeng Xiang, Yu Deng, Xin Tong, and Jiaolong Yang.
1030 Moge: Unlocking accurate monocular geometry estimation for open-domain images with optimal
1031 training supervision, 2024a. URL <https://arxiv.org/abs/2410.19115>.

1032 Shuzhe Wang, Vincent Leroy, Yohann Cabon, Boris Chidlovskii, and Jerome Revaud. Dust3r:
1033 Geometric 3d vision made easy. In *Proc. CVPR*, 2024b.

1034 Xi Wang, Robin Courant, Marc Christie, and Vicky Kalogeiton. Akira: Augmentation kit on rays for
1035 optical video generation. In *Proc. CVPR*, 2025c.

1036 Xiaodong Wang, Zhirong Wu, and Peixi Peng. Longdwm: Cross-granularity distillation for building
1037 a long-term driving world model. *arXiv preprint arXiv:2506.01546*, 2025d.

1038 Yikai Wang, Xinzhou Wang, Zilong Chen, Zhengyi Wang, Fuchun Sun, and Jun Zhu. Vidu4d: Single
1039 generated video to high-fidelity 4d reconstruction with dynamic gaussian surfels. *arXiv preprint*
1040 *arXiv:2405.16822*, 2024c.

1041 Yikai Wang, Guangce Liu, Xinzhou Wang, Zilong Chen, Jiafang Li, Xin Liang, Fuchun Sun, and
1042 Jun Zhu. Video4dgen: Enhancing video and 4d generation through mutual optimization. *arXiv*
1043 *preprint arXiv:2504.04153*, 2025e.

1044 Yizhi Wang, Mingrui Zhao, Ali Mahdavi-Amiri, and Hao Zhang. Act-r: Adaptive camera trajectories
1045 for single view 3d reconstruction. *arXiv preprint arXiv:2505.08239*, 2025f.

1046 Yuelei Wang, Jian Zhang, Pengtao Jiang, Hao Zhang, Jinwei Chen, and Bo Li. Cpa: Camera-
1047 pose-awareness diffusion transformer for video generation. *arXiv preprint arXiv:2412.01429*,
1048 2024d.

1049 Zhengyi Wang, Cheng Lu, Yikai Wang, Fan Bao, Chongxuan Li, Hang Su, and Jun Zhu. Prolific-
1050 Dreamer: High-fidelity and diverse text-to-3D generation with variational score distillation. In
1051 *Proc. NeurIPS*, 2023b.

1052 Zhouxia Wang, Ziyang Yuan, Xintao Wang, Tianshui Chen, Menghan Xia, Ping Luo, and Yin Shan.
1053 Motionctrl: A unified and flexible motion controller for video generation. In *SIGGRAPH*, 2024e.

1054 Daniel Watson, William Chan, Ricardo Martin-Brualla, Jonathan Ho, Andrea Tagliasacchi, and
1055 Mohammad Norouzi. Novel view synthesis with diffusion models. *Proc. ICLR*, 2023.

1056 Daniel Watson, Saurabh Saxena, Lala Li, Andrea Tagliasacchi, and David J Fleet. Controlling space
1057 and time with diffusion models. *arXiv preprint arXiv:2407.07860*, 2024.

1058 Qi Wu, Janick Martinez Esturo, Ashkan Mirzaei, Nicolas Moenne-Loccoz, and Zan Gojcic. 3dgt:
1059 Enabling distorted cameras and secondary rays in gaussian splatting. In *Proc. CVPR*, 2025a.

1060 Rundi Wu, Ruiqi Gao, Ben Poole, Alex Trevithick, Changxi Zheng, Jonathan T Barron, and Alek-
1061 sander Holynski. Cat4d: Create anything in 4d with multi-view video diffusion models. *Proc.*
1062 *CVPR*, 2025b.

1063 Tong Wu, Shuai Yang, Ryan Po, Yinghao Xu, Ziwei Liu, Dahua Lin, and Gordon Wetzstein. Video
1064 world models with long-term spatial memory. *arXiv preprint arXiv:2506.05284*, 2025c.

1065 Zijie Wu, Chaohui Yu, Yanqin Jiang, Chenjie Cao, Fan Wang, and Xiang Bai. Sc4d: Sparse-controlled
1066 video-to-4d generation and motion transfer. *arXiv preprint arXiv:2404.03736*, 2024.

1067 Zeqi Xiao, Yifan Zhou, Shuai Yang, and Xingang Pan. Video diffusion models are training-free
1068 motion interpreter and controller. *arXiv preprint arXiv:2405.14864*, 2024.

1069 Kevin Xie, Jonathan Lorraine, Tianshi Cao, Jun Gao, James Lucas, Antonio Torralba, Sanja Fidler,
1070 and Xiaohui Zeng. LATTE3D: Large-scale amortized text-to-enhanced3D synthesis. In *Proc.*
1071 *ECCV*, 2024a.

1080 Yiming Xie, Chun-Han Yao, Vikram Voleti, Huaizu Jiang, and Varun Jampani. Sv4d: Dynamic 3d
1081 content generation with multi-frame and multi-view consistency. *arXiv preprint arXiv:2407.17470*,
1082 2024b.

1083 Jinbo Xing, Long Mai, Cusuh Ham, Jiahui Huang, Aniruddha Mahapatra, Chi-Wing Fu, Tien-Tsin
1084 Wong, and Feng Liu. Motioncanvas: Cinematic shot design with controllable image-to-video
1085 generation. In *SIGGRAPH*, 2025.

1086 Dejia Xu, Yifan Jiang, Chen Huang, Liangchen Song, Thorsten Gernoth, Liangliang Cao, Zhangyang
1087 Wang, and Hao Tang. Cavia: Camera-controllable multi-view video diffusion with view-integrated
1088 attention. *arXiv preprint arXiv:2410.10774*, 2024a.

1089 Dejia Xu, Hanwen Liang, Neel P Bhatt, Hezhen Hu, Hanxue Liang, Konstantinos N Plataniotis,
1090 and Zhangyang Wang. Comp4d: Llm-guided compositional 4d scene generation. *arXiv preprint*
1091 *arXiv:2403.16993*, 2024b.

1092 Dejia Xu, Weili Nie, Chao Liu, Sifei Liu, Jan Kautz, Zhangyang Wang, and Arash Vahdat. Camco:
1093 Camera-controllable 3d-consistent image-to-video generation. *arXiv preprint arXiv:2406.02509*,
1094 2024c.

1095 Yinghao Xu, Zifan Shi, Wang Yifan, Hansheng Chen, Ceyuan Yang, Sida Peng, Yujun Shen, and
1096 Gordon Wetzstein. GRM: Large Gaussian reconstruction model for efficient 3D reconstruction
1097 and generation. In *Proc. ECCV*, 2024d.

1098 Yinghao Xu, Hao Tan, Fujun Luan, Sai Bi, Peng Wang, Jiahao Li, Zifan Shi, Kalyan Sunkavalli,
1099 Gordon Wetzstein, Zexiang Xu, et al. DMV3D: Denoising multi-view diffusion using 3D large
1100 reconstruction model. In *Proc. ICLR*, 2024e.

1101 Zhen Xu, Zhengqin Li, Zhao Dong, Xiaowei Zhou, Richard Newcombe, and Zhaoyang Lv.
1102 4dgt: Learning a 4d gaussian transformer using real-world monocular videos. *arXiv preprint*
1103 *arXiv:2506.08015*, 2025.

1104 Zhongcong Xu, Jianfeng Zhang, Jun Hao Liew, Wenqing Zhang, Song Bai, Jiashi Feng, and
1105 Mike Zheng Shou. Pv3d: A 3d generative model for portrait video generation. In *Proc. ICLR*,
1106 2023.

1107 Yunzhi Yan, Zhen Xu, Haotong Lin, Haian Jin, Haoyu Guo, Yida Wang, Kun Zhan, Xianpeng Lang,
1108 Hujun Bao, Xiaowei Zhou, et al. Streetcrafter: Street view synthesis with controllable video
1109 diffusion models. In *Proc. CVPR*, 2025.

1110 Qitong Yang, Mingtao Feng, Zijie Wu, Shijie Sun, Weisheng Dong, Yaonan Wang, and Ajmal Mian.
1111 Beyond skeletons: Integrative latent mapping for coherent 4d sequence generation. *arXiv preprint*
1112 *arXiv:2403.13238*, 2024a.

1113 Zeyu Yang, Zijie Pan, Chun Gu, and Li Zhang. Diffusion²: Dynamic 3d content generation via score
1114 composition of orthogonal diffusion models. *arXiv preprint 2404.02148*, 2024b.

1115 Zhongqi Yang, Wenhao Ge, Yuqi Li, Jiaqi Chen, Haoyuan Li, Mengyin An, Fei Kang, Hua Xue,
1116 Baixin Xu, Yuyang Yin, et al. Matrix-3d: Omnidirectional explorable 3d world generation. *arXiv*
1117 *preprint arXiv:2508.08086*, 2025.

1118 Junliang Ye, Fangfu Liu, Qixiu Li, Zhengyi Wang, Yikai Wang, Xinzhou Wang, Yueqi Duan,
1119 and Jun Zhu. DreamReward: Text-to-3D generation with human preference. *arXiv preprint*
1120 *arXiv:2403.14613*, 2024.

1121 Vickie Ye, Ruilong Li, Justin Kerr, Matias Turkulainen, Brent Yi, Zhuoyang Pan, Otto Seiskari,
1122 Jianbo Ye, Jeffrey Hu, Matthew Tancik, and Angjoo Kanazawa. gsplat: An open-source library for
1123 gaussian splatting. *JMLR*, 2025.

1124 Yuyang Yin, Dejia Xu, Zhangyang Wang, Yao Zhao, and Yunchao Wei. 4dgen: Grounded 4d content
1125 generation with spatial-temporal consistency. *arXiv preprint arXiv:2312.17225*, 2023.

1126 Paul Yoo, Jiaxian Guo, Yutaka Matsuo, and Shixiang Shane Gu. DreamSparse: Escaping from Plato’s
1127 cave with 2D diffusion model given sparse views. In *arXiv preprint arXiv:2306.03414*, 2023.

1134 Heng Yu, Chaoyang Wang, Peiye Zhuang, Willi Menapace, Aliaksandr Siarohin, Junli Cao, Laszlo A
1135 Jeni, Sergey Tulyakov, and Hsin-Ying Lee. 4real: Towards photorealistic 4d scene generation via
1136 video diffusion models. In *Proc. NeurIPS*, 2024a.

1137 Mark YU, Wenbo Hu, Jinbo Xing, and Ying Shan. Trajectorycrafter: Redirecting camera trajectory
1138 for monocular videos via diffusion models. *Proc. ICCV*, 2025.

1139 Wangbo Yu, Jinbo Xing, Li Yuan, Wenbo Hu, Xiaoyu Li, Zhipeng Huang, Xiangjun Gao, Tien-
1140 Tsin Wong, Ying Shan, and Yonghong Tian. Viewcrafter: Taming video diffusion models for
1141 high-fidelity novel view synthesis. *arXiv preprint arXiv:2409.02048*, 2024b.

1142 Xin Yu, Yuan-Chen Guo, Yangguang Li, Ding Liang, Song-Hai Zhang, and Xiaojuan Qi. Text-to-3D
1143 with classifier score distillation. *arXiv preprint arXiv:2310.19415*, 2023.

1144 Yu Yuan, Xijun Wang, Yichen Sheng, Prateek Chennuri, Xingguang Zhang, and Stanley Chan.
1145 Generative photography: Scene-consistent camera control for realistic text-to-image synthesis. In
1146 *Proc. CVPR*, 2025.

1147 Yu-Jie Yuan, Leif Kobbelt, Jiwen Liu, Yuan Zhang, Pengfei Wan, Yu-Kun Lai, and Lin Gao. 4dynamic:
1148 Text-to-4d generation with hybrid priors. *arXiv preprint arXiv:2407.12684*, 2024.

1149 Bohan Zeng, Ling Yang, Siyu Li, Jiaming Liu, Zixiang Zhang, Juanxi Tian, Kaixin Zhu, Yongzhen
1150 Guo, Fu-Yun Wang, Minkai Xu, et al. Trans4d: Realistic geometry-aware transition for composi-
1151 tional text-to-4d synthesis. *arXiv preprint arXiv:2410.07155*, 2024a.

1152 Yifei Zeng, Yanqin Jiang, Siyu Zhu, Yuanxun Lu, Youtian Lin, Hao Zhu, Weiming Hu, Xun
1153 Cao, and Yao Yao. Stag4d: Spatial-temporal anchored generative 4d gaussians. *arXiv preprint*
1154 *arXiv:2403.14939*, 2024b.

1155 Bowen Zhang, Tianyu Yang, Yu Li, Lei Zhang, and Xi Zhao. Compress3D: a compressed latent
1156 space for 3D generation from a single image. *arXiv preprint arXiv:2403.13524*, 2024a.

1157 David Junhao Zhang, Roni Paiss, Shiran Zada, Nikhil Karnad, David E Jacobs, Yael Pritch, Inbar
1158 Mosseri, Mike Zheng Shou, Neal Wadhwa, and Nataniel Ruiz. Recapture: Generative video camera
1159 controls for user-provided videos using masked video fine-tuning. *arXiv preprint arXiv:2411.05003*,
1160 2024b.

1161 Haiyu Zhang, Xinyuan Chen, Yaohui Wang, Xihui Liu, Yunhong Wang, and Yu Qiao. 4diffusion:
1162 Multi-view video diffusion model for 4d generation. *arXiv preprint arXiv:2405.20674*, 2024c.

1163 Hao Zhang, Di Chang, Fang Li, Mohammad Soleymani, and Narendra Ahuja. Magicpose4d: Crafting
1164 articulated models with appearance and motion control. *arXiv preprint arXiv:2405.14017*, 2024d.

1165 Kai Zhang, Sai Bi, Hao Tan, Yuanbo Xiangli, Nanxuan Zhao, Kalyan Sunkavalli, and Zexiang Xu.
1166 Gs-lrm: Large reconstruction model for 3d gaussian splatting. In *Proc. ECCV*, 2024e.

1167 Richard Zhang, Phillip Isola, Alexei A Efros, Eli Shechtman, and Oliver Wang. The unreasonable
1168 effectiveness of deep features as a perceptual metric. In *Proc. CVPR*, 2018.

1169 Songchun Zhang, Huiyao Xu, Sitong Guo, Zhongwei Xie, Hujun Bao, Weiwei Xu, and Changqing
1170 Zou. Spatialcrafter: Unleashing the imagination of video diffusion models for scene reconstruction
1171 from limited observations. *arXiv preprint arXiv:2505.11992*, 2025a.

1172 Tianyuan Zhang, Hong-Xing Yu, Rundui Wu, Brandon Y Feng, Changxi Zheng, Noah Snavely, Jiajun
1173 Wu, and William T Freeman. Physdreamer: Physics-based interaction with 3d objects via video
1174 generation. In *Proc. ECCV*, 2024f.

1175 Yisu Zhang, Chenjie Cao, Chaohui Yu, and Jianke Zhu. Lion-lora: Rethinking lora fusion to unify
1176 controllable spatial and temporal generation for video diffusion. *arXiv preprint arXiv:2507.05678*,
1177 2025b.

1178 Zhichao Zhang, Hui Chen, Jinsheng Deng, Xiaoqing Yin, Xingshen Song, and Ming Xu. Motion4d:
1179 A decoupled pipeline for enhanced text-to-4d generation with optimized motion patterns. *SSRN*,
1180 2024g.

1188 Yuyang Zhao, Zhiwen Yan, Enze Xie, Lanqing Hong, Zhenguo Li, and Gim Hee Lee. Animate124:
1189 Animating one image to 4d dynamic scene. *arXiv preprint arXiv:2311.14603*, 2023.

1190

1191 Yuyang Zhao, Chung-Ching Lin, Kevin Lin, Zhiwen Yan, Linjie Li, Zhengyuan Yang, Jianfeng
1192 Wang, Gim Hee Lee, and Lijuan Wang. Genxd: Generating any 3d and 4d scenes. *arXiv preprint*
1193 *arXiv:2411.02319*, 2024.

1194 Guangcong Zheng, Teng Li, Rui Jiang, Yehao Lu, Tao Wu, and Xi Li. Cami2v: Camera-controlled
1195 image-to-video diffusion model. *arXiv preprint arXiv:2410.15957*, 2024a.

1196

1197 Yufeng Zheng, Xueting Li, Koki Nagano, Sifei Liu, Otmar Hilliges, and Shalini De Mello. A unified
1198 approach for text-and image-guided 4d scene generation. In *Proc. CVPR*, 2024b.

1199 Junwei Zhou, Xueting Li, Lu Qi, and Ming-Hsuan Yang. Coco4d: Comprehensive and complex 4d
1200 scene generation. *arXiv preprint arXiv:2506.19798*, 2025.

1201

1202 Tinghui Zhou, Richard Tucker, John Flynn, Graham Fyffe, and Noah Snavely. Stereo magnification:
1203 Learning view synthesis using multiplane images. In *SIGGRAPH*, 2018.

1204 Hanxin Zhu, Tianyu He, Anni Tang, Junliang Guo, Zhibo Chen, and Jiang Bian. Compositional
1205 3d-aware video generation with llm director. *arXiv preprint arXiv:2409.00558*, 2024.

1206

1207 Hanxin Zhu, Tianyu He, Xiqian Yu, Junliang Guo, Zhibo Chen, and Jiang Bian. Ar4d: Autoregressive
1208 4d generation from monocular videos. *arXiv preprint arXiv:2501.01722*, 2025.

1209 Chen Ziwen, Hao Tan, Kai Zhang, Sai Bi, Fujun Luan, Yicong Hong, Li Fuxin, and Zexiang Xu.
1210 Long-lrm: Long-sequence large reconstruction model for wide-coverage gaussian splats. *arXiv*
1211 *preprint arXiv:2410.12781*, 2024.

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A ADDITIONAL DETAILS OF VIDEO DIFFUSION MODEL

This section provides additional detail on the GEN3C (Ren et al., 2025) video diffusion model backbone introduced in Sec. 3.1 and on how we further improve its 3D consistency, which underpins the teacher quality our first contribution depends on.

Conservative mask refinement. A key limitation of GEN3C’s forward warping is background leakage: occluded object parts in the source view (e.g., the lion body in Fig. 8) become visible in novel viewpoints but are not flagged in the disocclusion mask. The result is incomplete foreground completion by the video diffusion model, as shown in Fig. 8 (second and fourth columns).

To address this, we adopt a conservative strategy: because the geometry of occluded regions in $\mathbf{P}^{t,v}$ is unknown, we assume all such areas are occupied and derive refined disocclusion masks accordingly, letting the video model reason about potentially visible foreground regions missed by standard point-based rendering.

Our approach constructs a triangular mesh from camera-space points $\mathbf{P}_c^{t,v} = \mathbf{C}^t \mathbf{P}^{t,v}$ by connecting spatially adjacent pixels, similar to Hu et al. (2021):

$$\mathcal{M} = \bigcup_{(u,v)} \{ \triangle(\mathbf{p}_{u,v}, \mathbf{p}_{u+1,v}, \mathbf{p}_{u,v+1}), \triangle(\mathbf{p}_{u+1,v}, \mathbf{p}_{u+1,v+1}, \mathbf{p}_{u,v+1}) \} \quad (2)$$

where $\mathbf{p}_{u,v}$ denotes the 3D point at pixel (u, v) in $\mathbf{P}_c^{t,v}$. This surface mesh serves as a boundary between visible and invisible regions. For each rendered pixel, we compare the standard point-based depth $\mathbf{D}^{t,v}$ with the mesh-interpolated depth $\mathbf{D}_{\mathcal{M}}^{t,v}$ obtained via ray-surface intersection:

$$\mathbf{M}^{t,v}(u, v) = \begin{cases} 0 & \text{if } \mathbf{D}_{\mathcal{M}}^{t,v}(u, v) < \mathbf{D}^{t,v}(u, v) - \epsilon, \\ \mathbf{M}_{\text{orig}}^{t,v}(u, v) & \text{otherwise} \end{cases}, \quad (3)$$

where $\mathbf{M}_{\text{orig}}^{t,v}$ denotes the original disocclusion mask from forward warping and ϵ is a small tolerance factor. When the mesh surface is closer than the original points, we conservatively mask out these regions as potential foreground disocclusions. This produces more reliable guidance for video generation, especially under large viewpoint changes, as shown in Fig. 8 (third and fifth columns).

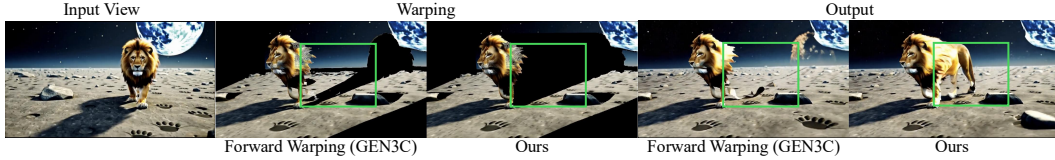


Figure 8: **Rendering function comparison.** Our model builds upon GEN3C (Ren et al., 2025) that uses forward warped images as camera control conditioning. We visualize forward warping (left) vs. our conservative rendering function (right). The highlighted region is incorrectly filled with background pixels in forward warping but properly masked in ours, enabling correct completion.

B MODEL DETAILS

Progressive training. Directly training the 3DGS decoder on 6 trajectories of 121 frames at 704×1280 is expensive, so we train progressively, as shown in Tab. 3; the dynamic model of our second contribution is then initialized from the final static model. Total training takes 6 days on 8 NVIDIA A100 80 GB GPUs, and we use gsplat (Ye et al., 2025) as the 3DGS implementation.

Subsampling Gaussians. Generating pixel-aligned 3D Gaussians scales linearly with spatial and temporal resolution, so it is a natural place to control cost. Our generated scenes span 726 frames at 704×1280 , which would amount to 654,213,120 per-pixel Gaussians. We therefore emit one 3D Gaussian per 8×8 spatial neighborhood, cutting the count by a factor of 64 to 10,222,080, and our opacity-based pruning further reduces it to 2,044,416, yielding a compact output.

Joint Transformer-Mamba blocks. We use the reconstruction block design of Long-LRM (Ziwen et al., 2024): one block consists of one Transformer layer followed by seven Mamba-2 layers, repeated

Table 3: **Progressive training setup.** Stages 1–6 build up the static 3DGS decoder of our first contribution, and stage 7 fine-tunes it for the dynamic extension of our second contribution.

Stage	$H \times W$	L	V	S	B	Steps
Static						
1	176×320	17	1	17	4	10k
2	176×320	49	1	49	4	2.5k
3	352×640	49	1	49	2	2.5k
4	704×1280	49	1	49	1	2.5k
5	704×1280	121	1	9	1	57.5k
6	704×1280	121	1–6	9	1	7k
Dynamic						
7	704×1280	121	6	12	1	10k

twice to give 16 layers with 512 hidden dimensions. Notably, this network is much smaller than those of Long-LRM (Ziwen et al., 2024) or Wonderland (Liang et al., 2025a), which use 24 layers with 1024 hidden dimensions — an efficiency that reinforces the scalability argument behind our decoder. While pure Transformers are typically more powerful, they are far slower to train and run than Mamba-2. Because we trained both configurations under the same budget, we attribute the Transformer-only variant’s lower rendering quality to needing considerably more iterations to catch up.

Camera encoding. Following prior work (Kant et al., 2024; Zhang et al., 2024e), we use Plücker embeddings for camera encoding, computed at full spatial and temporal resolution and projected into the video latent space with the pre-trained RGB encoder. Because that encoder expects 3 input channels but Plücker embeddings are 6-dimensional, we separately encode the 3-dimensional ray directions and the 3-dimensional cross product of ray directions and origins, then concatenate the resulting latents along the channel dimension — reusing the pre-trained encoder rather than adding a new one.

Time encoding. The time encoding for our dynamic extension follows the same reuse principle. To make the 1-dimensional time input compatible with the 3-channel RGB encoder, we concatenate a 2-dimensional sinusoidal embedding to the original time value along the channel dimension, replicate these values across the spatial dimensions, and then encode them into the video latent space. This lets us independently encode both the source and target times into the compressed latent representation.

Dynamic data augmentation. Figure 9 visualizes our dynamic data augmentation for two example camera trajectories. The procedure creates pairs of supervision views that cover the scene from the same motion state but from opposite extreme viewpoints, which is what prevents geometry collapse and makes our dynamic-scene contribution robust.

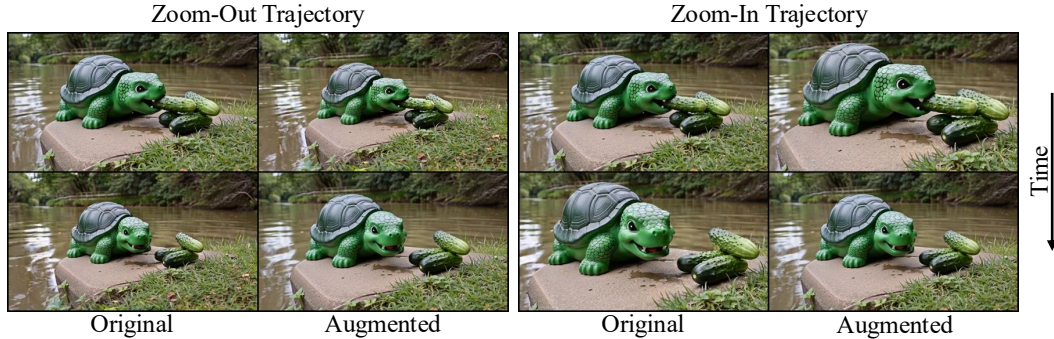


Figure 9: **Dynamic data augmentation videos.** We augment the supervision data with a motion-reversed video, ensuring that each timestep is observed from the full spatial coverage, thereby preventing low opacity artifacts in the early timesteps. We show two example trajectories, i.e., zoom-out and zoom-in, and visualize their corresponding augmented videos. The augmented videos are flipped in their camera motion.



Figure 10: **Image-to-3DGS Comparison.** We compare our method with BTimer (GEN3C) and visualize five views from generated scenes. We observe significantly fewer artifacts and higher fidelity.

C ADDITIONAL EVALUATION

C.1 ADDITIONAL BASELINE DETAILS

To further stress-test our third contribution, we additionally compare against BTimer (Liang et al., 2025b), a recent approach for joint 3D and 4D reconstruction from posed images or videos. We reached out to the BTimer authors, who ran experiments for us on the same scenes as *Lyra*. Since BTimer is purely regression-based, we integrate it with our GEN3C (Ren et al., 2025) video diffusion backbone, using the same camera trajectories as our model and feeding RGB-decoded GEN3C videos as input. BTimer operates in pixel space and requires 12 input images; as noted in the main paper, such pixel-space models run out of memory on the 726 high-resolution frames GEN3C produces. We therefore uniformly subsample GEN3C frames: for static evaluations we select 12 frames that evenly cover the viewpoints, and for dynamic evaluations we sample 11 frames spanning the viewpoint range plus the bullet-time frame for the motion being reconstructed, so the target motion state is always present as a reference. We denote this integrated baseline BTimer (GEN3C), i.e., BTimer combined with GEN3C.

C.2 ADDITIONAL STATIC 3D EVALUATION

For image-to-3DGS generation, Fig. 10 shows qualitative comparisons with BTimer (GEN3C) and Tab. 4a the corresponding quantitative results. Both confirm the same conclusion as the main paper: our method yields higher quality with fewer artifacts.

C.3 DYNAMIC 3D EVALUATION

BTimer (GEN3C) comparison. For dynamic 3D evaluation, we compare against BTimer (GEN3C) (Sec. C.1) on 100 out-of-distribution videos from our dynamic *Lyra* dataset, cropping to 512×512 for a fair comparison since BTimer was trained mainly at that resolution. As reported in Tab. 4b, our method significantly outperforms BTimer (GEN3C), extending our state-of-the-art contribution to the dynamic setting.

Table 4: **Quantitative results on static and dynamic *Lyra* datasets.** Together, the two panels support our third contribution across both the static and dynamic regimes against the BTimer (GEN3C) baseline.

Method	PSNR $\uparrow$	SSIM $\uparrow$	LPIPS $\downarrow$	Method	PSNR $\uparrow$	SSIM $\uparrow$	LPIPS $\downarrow$
BTimer (GEN3C)	16.32	0.580	0.427	BTimer (GEN3C)	20.29	0.687	0.315
Ours	24.92	0.834	0.183	Ours	23.07	0.779	0.231

(a) **Static *Lyra* dataset.** Our method substantially outperforms the baseline. (b) **Dynamic *Lyra* dataset.** Our 4D extension clearly leads the baseline.

C.4 DEPTH VISUALIZATION

Figure 11 visualizes depths from our generated 3DGS. As claimed in the ablations, depth supervision prevents flat geometry without sacrificing visual quality.

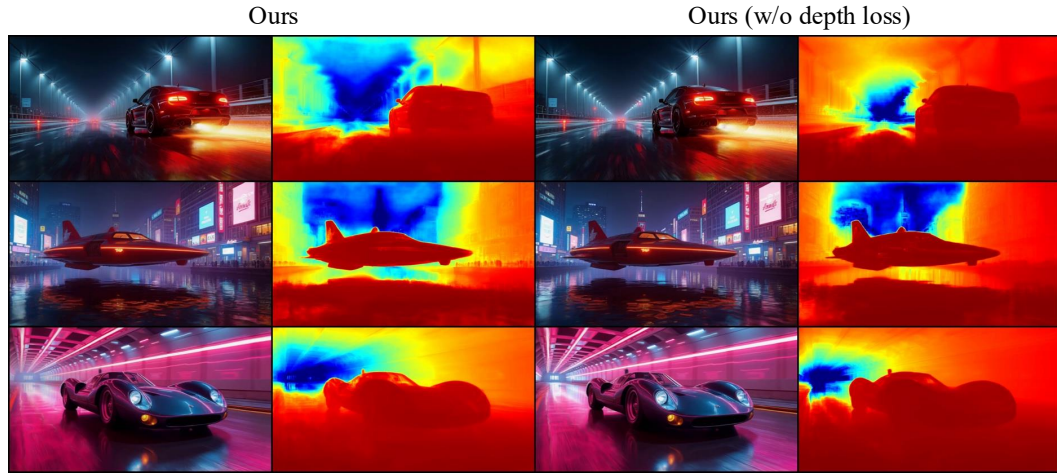


Figure 11: **Depth loss ablation.** We visualize 3DGS renderings and corresponding depth maps. Using the depth loss as additional supervision prevents flat geometries without sacrificing visual quality.

D ROBOT SIMULATION

A primary motivation for building a diverse 3D generator is synthesizing simulation environments for autonomous agents, and this section shows that our contributions translate into a working downstream pipeline. End to end, we generate 3D scenes from text with our model, export the 3D Gaussians as a .ply file, and convert them to .usdz using 3DGUT (Wu et al., 2025a) and its export function. The resulting .usdz can be imported into the NVIDIA Isaac robot development platform as a physically based virtual environment in which AI-based robots can be trained and tested under diverse conditions. Early demo results will be released upon acceptance; a screenshot of the interface is shown in Fig. 12.

E RELATED WORK

This section discusses additional related work and, throughout, positions our contributions against it. As a visual summary, Fig. 13 highlights how our approach differs from prior works such as CAT3D (Gao et al., 2024b) and Bolt3D (Szymanowicz et al., 2025b).

3D generation. Early approaches to 3D generation primarily targeted single object categories, extending GANs into 3D by using neural renderers as an inductive bias (DeVries et al., 2021; Chan et al., 2022; Or-El et al., 2022; Schwarz et al., 2022; Bahmani et al., 2023b). CLIP-based supervision (Radford et al., 2021) then broadened the field toward more flexible, diverse asset creation with text-driven generation and editing (Chen et al., 2018; Jain et al., 2022; Sanghi et al., 2022;

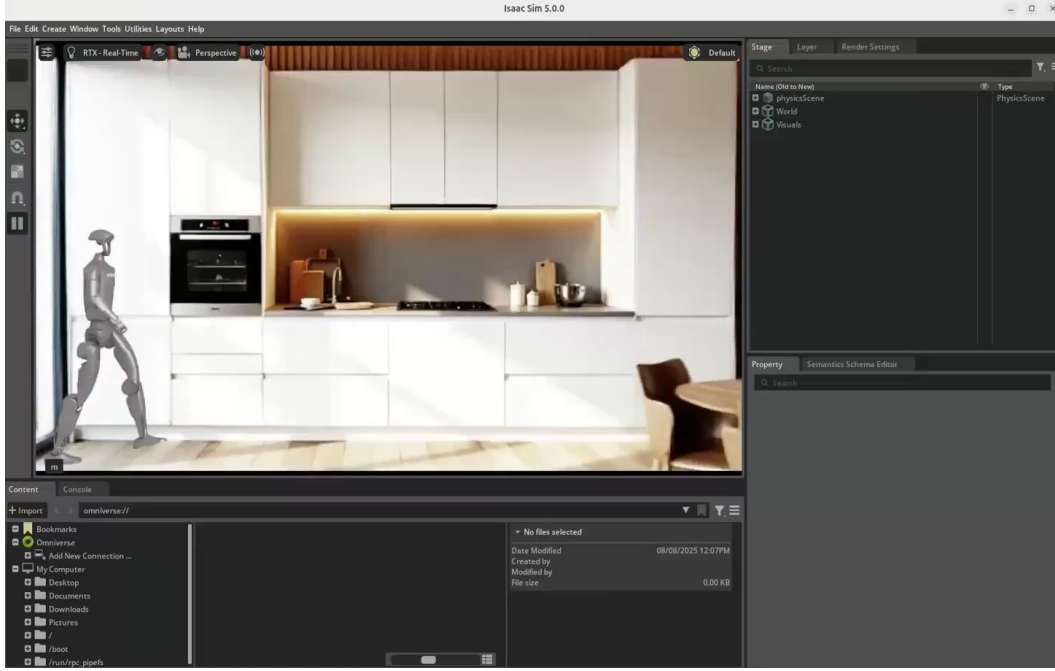


Figure 12: **Robot Simulation.** We visualize a frame of a robot simulation within the Isaac Sim 5.0 simulation framework that takes a generated 3DGS scene from our method as input.

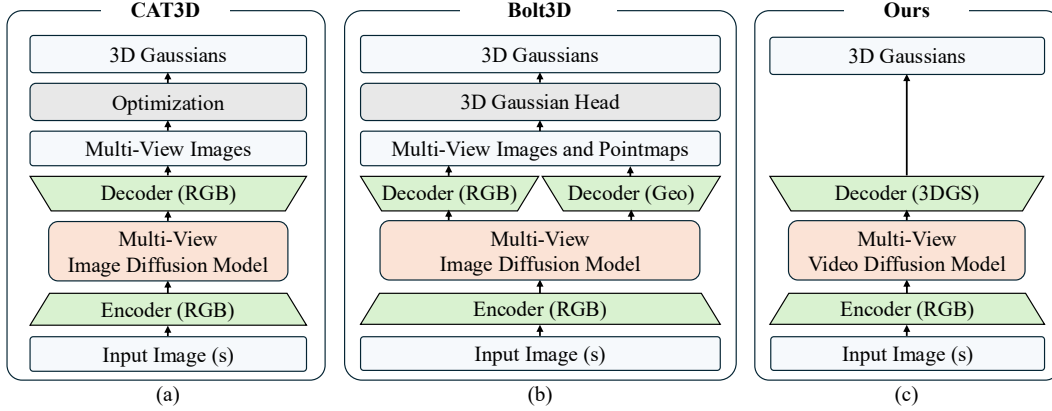


Figure 13: **Approaches for 3D Generation.** CAT3D (Gao et al., 2024b) proposed a multi-view image diffusion model that outputs images from novel viewpoints; subsequently, the images are reconstructed into 3D Gaussians using optimization. Bolt3D (Szymanowicz et al., 2025b) fine-tunes CAT3D to output pointmaps using a geometry autoencoder; instead of relying on optimization, the 3D scene is predicted with a feed-forward 3D Gaussian head. In contrast, our work builds upon a pre-trained camera-controlled video diffusion model and *directly* decodes the multi-view video latents into 3D Gaussians.

Jetchev, 2021; Gao et al., 2023; Wang et al., 2022), and diffusion models later improved quality further by replacing CLIP guidance with Score Distillation Sampling (SDS) (Poole et al., 2023; Wang et al., 2023b; Lin et al., 2023a; Chen et al., 2023b; Liang et al., 2023; Wang et al., 2023a; Li et al., 2024e; He et al., 2024b; Ye et al., 2024; Liu et al., 2024b; Yu et al., 2023; Katzir et al., 2024; Lee et al., 2024a; Sun et al., 2024a). Unlike this line, our first contribution avoids per-scene SDS optimization entirely and instead distills a video model in a single feed-forward pass.

To further enhance structural coherence, several methods enforce multi-view consistency by generating scenes from different perspectives (Lin et al., 2023b; Liu et al., 2023; Shi et al., 2024; Feng et al., 2024a; Liu et al., 2024a; Kim et al., 2023; Voleti et al., 2024; Höllein et al., 2024; Tang et al., 2024c;

Gao et al., 2024b; Wang et al., 2025f; Kant et al., 2025; Yuan et al., 2025; Ren et al., 2024b), while others adopt iterative inpainting as a strategy for scene synthesis (Höllein et al., 2023; Shriram et al., 2024). Another line of work explores lifting 2D images into 3D using NeRF (Mildenhall et al., 2020), 3D Gaussian Splatting (Kerbl et al., 2023), or mesh-based representations coupled with diffusion models (Chan et al., 2023; Tang et al., 2023a; Gu et al., 2023; Liu et al., 2024c; Yoo et al., 2023; Tewari et al., 2023; Qian et al., 2024b; Long et al., 2024; Wan et al., 2024; Szymanowicz et al., 2023; Lu et al., 2024).

Feed-forward 3D models. Complementary efforts investigate fast, feed-forward techniques that directly predict 3D content from images or text (Hong et al., 2024; Li et al., 2024b; Xu et al., 2024e;d; Zhang et al., 2024a; Han et al., 2024; Jiang & Wang, 2024; Xie et al., 2024a; Tang et al., 2024b; Tochilkin et al., 2024; Qian et al., 2024a; Szymanowicz et al., 2024; 2025a; Liang et al., 2025a; Szymanowicz et al., 2025b; Schwarz et al., 2025; Yang et al., 2025; Zhang et al., 2025a). In contrast to our second contribution, these approaches remain limited to static 3D scenes. Others target specific domains such as faces (Kirschstein et al., 2025), and concurrent work (Liang et al., 2025b; Xu et al., 2025) handles real scenes but not the diverse generated scenes or large viewpoint changes that our framework addresses.

4D generation. Recent years have seen rapid progress in 4D generation, i.e., dynamic 3D scene synthesis, with many approaches guided by input text prompts or images. Following the introduction of large-scale generative models for this task (Singer et al., 2023), subsequent works advanced both visual fidelity and motion quality (Ren et al., 2023; Ling et al., 2024a; Bahmani et al., 2024b; Zheng et al., 2024b; Gao et al., 2024a; Yang et al., 2024a; Jiang et al., 2024b; Miao et al., 2024; Li et al., 2024a; Zhang et al., 2024g; Yuan et al., 2024; Jiang et al., 2024a). Our second contribution sits in this space but is distinguished by producing an explicit, feed-forward 4D representation from monocular video.

While text conditioning is widely used, alternative methods aim to lift 2D images or videos into dynamic 3D scenes (Ren et al., 2023; Zhao et al., 2023; Yin et al., 2023; Pan et al., 2024; Zheng et al., 2024b; Ling et al., 2024a; Gao et al., 2024a; Zeng et al., 2024b; Chu et al., 2024; Wu et al., 2024; Yang et al., 2024b; Wang et al., 2024c; Feng et al., 2024b; Sun et al., 2024c; Zhang et al., 2024c; Ren et al., 2024a; Lee et al., 2024b; Li et al., 2024c; Van Hoorick et al., 2024; Uzolas et al., 2024; Chai et al., 2024; Liang et al., 2024; Li et al., 2024d; Xie et al., 2024b; Li et al., 2025a; Wang et al., 2025e; Zhu et al., 2025; Zhou et al., 2025; Hu et al., 2025; Park et al., 2025). Other efforts explore incorporating physics priors into generation pipelines (Lin et al., 2024; Huang et al., 2024b; Zhang et al., 2024f; Kiray et al., 2025), or enhancing motion controllability with template-driven approaches (Zhang et al., 2024d; Sun et al., 2024b; Chen et al., 2024b). A parallel direction emphasizes compositional and interactive 4D generation (Bahmani et al., 2024a; Xu et al., 2024b; Cao et al., 2024; Yu et al., 2024a; Zeng et al., 2024a; Zhu et al., 2024).

Another strand of research extends 3D GAN frameworks into the 4D setting by training directly on 2D video data (Bahmani et al., 2023a; Xu et al., 2023). Such methods are typically limited by small-scale, single-category datasets that constrain generalization and remain object-centric, often neglecting backgrounds. Consequently, their visual fidelity lags behind the photorealism of recent video generation models — precisely the photorealism our first contribution leverages by distilling from such a model rather than competing with it.

Camera-conditioned video models. Recent progress has focused on incorporating camera control into video diffusion models. The pioneering work MotionCtrl (Wang et al., 2024e) introduced camera conditioning by augmenting pre-trained video models (Chen et al., 2023a; Blattmann et al., 2023) with extrinsic matrices. Subsequent efforts (He et al., 2024a; Xu et al., 2024c; Kuang et al., 2024; Ju et al., 2025; He et al., 2025; Li et al., 2025c; Wang et al., 2025c; 2024d; Lei et al., 2025; Liu et al., 2025) enhanced conditioning by representing cameras using Plücker coordinates, while others use follow MotionCtrl to simply use poses flattened (Wang et al., 2025b). Another line of research (Hu et al., 2024; Xiao et al., 2024; Ling et al., 2024c; Hou et al., 2024) enabled camera motion control without introducing additional trainable parameters. Despite these advances, all of the above rely on U-Net-based architectures. Other approaches use low-rank approaches for camera control (Zhang et al., 2025b).

Several recent works continue to extend U-Net-based approaches for camera control (Zhao et al., 2024; Xu et al., 2024a; Zheng et al., 2024a; Zhang et al., 2024b; Yu et al., 2024b; Lei et al., 2024), while

1566 others begin exploring diffusion transformers. Cheong et al.(Cheong et al., 2024) propose one such
1567 transformer-based framework, though its scene and visual quality remain limited. DimensionX(Sun
1568 et al., 2024d) introduces joint space-time control in diffusion transformers, but relies on pre-defined,
1569 non-continuous camera trajectories. Other works (Huang et al., 2024a; Mai et al., 2025; Jiang et al.,
1570 2025b) instead explore pose and geometry estimation with a video DiT. CAT4D (Wu et al., 2025b)
1571 proposes a multi-view video diffusion model derived from fine-tuning a multi-view backbone. Large-
1572 scale synthetic data generation further boosts performance, as demonstrated by SynCamMaster (Bai
1573 et al., 2025b) and ReCamMaster (Bai et al., 2025a), which achieve strong results on camera-controlled
1574 video generation using transformer-based architectures. Finally, 4DiM (Watson et al., 2024) trains a
1575 space-time diffusion model from scratch for novel view synthesis from a single input image. Another
1576 line of work edits camera movements of videos with underlying motion using camera retargeting (Seo
1577 et al., 2025; Ma et al., 2025). One popular direction has been to use a spatio-temporal cache, e.g.,
1578 using point clouds or optical flow, to condition the camera-controlled video generation (Wu et al.,
1579 2025c; Ren et al., 2025; Wang et al., 2025d; Cao et al., 2025; YU et al., 2025; Jin et al., 2025; Xing
1580 et al., 2025; Yan et al., 2025; Li et al., 2025b; Gu et al., 2025).